

The Value of Names - Civil Society, Information, and Governing Multinationals on the Global Periphery*

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Abstract

Globally, civil society plays a crucial role in governance where laws and authority are weak. Increasingly, multinationals operate in these complex environments. We study how a key strategy of international civil society—disseminating information about human rights abuse—impacts multinationals. To do so, we focus on trend at the center international campaigns: the assassination of activists, and collect 20 years of murders related to the global mining sector. Using event study methodology, we estimate the impact human rights reporting on the stock price of firms connected to events. We find that the effect of the human rights spotlight is substantial. Firms named in assassination coverage have large, persistent, negative abnormal returns following assassinations. We calculate a median loss in market capitalisation of 100 million USD. We show media plays a crucial role; the negative impact of assassinations are strongest when they coincide with calm news cycles versus peak news cycles, when they may be crowded out by large, international stories. In addition, we argue our results are driven by events where companies are explicitly named in the reporting. Last, we show activist assassinations are positively related to the royalties paid by mining projects to domestic governments.

1 Introduction

Multinational corporations are political institutions. By revenue alone, the largest global corporations rival the size of nations.¹ As political actors, multinationals operate in a world without a single authority. Often where laws are murky and states are weak. For social scientists, the rising political power of multinationals presents a key problem of global governance (Fukuyama, 2016; Ruggie, 2018). Human rights scholars argue civil society may be indispensable in holding powerful, international actors accountable (Acemoglu and Robinson, 2020). Yet without formal

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¹Scholars have long juxtaposed multinational and states (Greene, 1983). Common means of doing so include revenue (Zingales, 2017) or market capitalisation. According to Global Justice Now (2018), in terms of revenue, 69 of the 100 largest economic entities are global corporations.

power, how does civil society govern? An influential scholarship argues that information and publicity are key weapons used by international activists to confront powerful agents (Brysk, 1993; Keck and Sikkink, 1998, 1999; Khagram et al., 2002).

The global natural resource sector illustrates the tension between civil society and multinationals. Home to some of the largest international firms, the sector is a flashpoint in the developing world. Since 2002, more global environmental activists have been killed than Australian or U.K. soldiers in war zones (Butt et al., 2019). Figure 2 shows the rise in assassination of activists surrounding mining activity between 1998 and 2019. These abuses—and those involved—are the focus of global publicity campaigns by human rights groups and media. Although a vibrant empirical scholarship has begun to explore how these strategies impact nations and formal political actors (Meernik et al., 2012; Hill Jr et al., 2013; Murdie and Peksen, 2014), we know little about their impact on private business. In the face of growing violence, how does the human rights spotlight effect sprawling multinationals?

We study the power of publicity, and its place in governing global firms. We study how the informational strategies employed by civil society (*e.g.* “naming-and-shaming” in the media) impacts multinational corporations. To answer this question, we turn to salient, well-publicized events at the heart of current advocacy: the assassinations of environmental activists. We focus on the sector at the center of this conflict, the global mining sector.

We estimate how publicity surrounding activist assassinations impact the stock prices of multinationals. Specifically, the mining companies—and their operations—named in international media coverage of these events. To do so we collect and code 20 years of data on activist assassination across the developing world. Many of these events involved conflict surrounding mining concessions and activity. Thus, we draw from hundreds of these high-profile incidents to identify the mining projects associated with these events and hand-match them to publicly-listed corporations. Doing so allows us to explore how markets respond to news of violent events surrounding their operations.

We estimate these effects using event study analysis. More precisely, we use two types of event studies to identify how publicity around human rights events are incorporated into the price of firms; each uses different variation from our data. We first 1) implement a traditional event study to estimate the abnormal returns of companies connected to assassinations on the days before and after an assassination. We then 2) build on our classic event study using the rich variation of our data. We estimate the impact of human rights publicity by comparing the abnormal returns for “associated” companies relative to control companies. Namely, control firms operating in the same sector, event-country, and event period. Both approaches allow us estimate the causal impact of publicity from different vantage points. Together, they show a robust pattern.

Our results show that high-profile human rights abuse has significant negative effects on firms caught in the spotlight. Following assassinations, we see significant, negative abnormal returns for profiled multinationals. These negative effects appear the day after a killing and grow steadily ten days after. For our classic event study, our findings are robust across a number test statistics, including new non-parametric tests. Our regression estimates show a similar, strong pattern. Companies named in assassination coverage have significant, negative abnormal returns following the events—effects which accumulate through time. Across both traditional and regression-based

event studies, we estimate similar negative impacts of being connected to human rights abuses through publicity. On days leading up to assassinations, we find no evidence of abnormal returns for these companies.

We interpret our results as showing that information disclosed around high-profile human rights news matters for markets. In fact, this negative impact is economically significant. For companies connected to events through publicity, we estimate that the median 10-day cumulative loss in market capitalisation is over 100 million USD. In other words, the informational strategies of international civil society impacts the bottom line of multinationals connected to the killing of activists.

Our study explores the underlying mechanisms in three ways.

First, we highlight the importance of media in driving the negative impacts. We examine this channel by considering the likelihood that news of human rights events reaches financial decision makers. More precisely, we ask how the impact of human rights events vary over the news cycle? We use daily “news pressure” data (Eisensee and Strömberg, 2007), and compare market reactions during periods with many newsworthy events to those with fewer newsworthy events. We find that the penalty of human rights news disappears when they coincide with more active news cycles. On other hand, the penalty survives when events occur during less eventful news periods—when news is less likely to be crowded out.

Second, we show large, informationally-sensitive investors react most to assassination events. That is, institutional investors that follow event-based trading strategies—such as hedge-funds—divest in mining companies following high-profile assassinations. Third, we highlight civil society’s informational strategy of “naming-and-shaming” in the media. We do so by comparing company named in connection to assassinations versus those merely operating close (geographically proximate) to events. We find that firms operations in the vicinity of events, though *not* named in media coverage, do not face a significant penalty, while firms named in the media do.

Last, we analyze the political economy surrounding our assassination events, and touch on the limits to governance in our setting. First, we do not find evidence that insiders trade off of prior knowledge of assassination events, inspired by Dube et al. (2011) analysis of coups. On the other hand, we show a relationship between domestic mining activity and the structure of local public finance. Assembling international data on mining royalties and government, we show that assassinations are positively correlated with royalties paid to domestic governments. Together, both findings indicate that downstream multinationals may not have full control over the costly actions local affiliates. Especially where local operations may collude with governments and paramilitary forces on periphery.

We contribute to a rich literature in empirical political economy and forensic economics. In particular, recent work using asset prices to understand how markets and international political events. In method and spirit, our work relate to large(r) sample event studies of global political shocks (Dube et al., 2011; Girardi, 2020).² Girardi (2020) uses a large sample of international election events to study how internal asset prices respond to formal political shocks across democracies. Using the sample of CIA-authorized coups, Dube et al. (2011) explore how the market responded

²More broadly, a number careful event studies explore how firm and sectors respond to formal political transitions. Including important examples in the UK (Herron, 2000) and US (Knight, 2006; Wagner et al., 2018).

favorably to US multinationals who stood to benefit for US coup authorisations. Our results stand on these papers, and suggest the use of asset prices to study firms as global political institutions unto themselves.

Our results suggest that market participants expect reputational costs in combination with disruptions from escalating conflict to outweigh potential gains from malfeasance. In doing so, we build on seminal forensic analysis on how firm assets respond to conflict in the developing world. Focusing on Egypt's Arab Spring, Acemoglu et al. (2017) finds returns fall significantly on days of protests for companies tied to the incumbent government, as investors adjust their expectations about their future rent-seeking ability.³ Guidolin and La Ferrara (2007) shows international diamond firms reacted negatively to deescalation of conflict in Angola. More broadly, Guidolin and La Ferrara (2010) document positive stock market reactions to the onset of conflict. DellaVigna and La Ferrara (2010) use similar methods to detect illegal arms trades in countries under arms embargo, and find that companies headquartered in corrupt countries may profit from increased conflict intensity. Our work suggests that the tools of civil society may help diminish returns to corporate misbehavior in the developing world.

Our results highlight the link between civil society and media—which we see as intertwined. As such, we contribute to a growing literature on the political economy of the media. In particular, research emphasizing the role of media in intermediation. Work by Miller (2006) and Dyck et al. (2010), stresses the importance of media in uncovering corporate fraud and promoting corporate accountability. Specifically, the impact of headline-grabbing cases that appeal to a broad audience and provide career incentives. The media's role as a monitor is furthermore illustrated by the effect of public scrutiny and negative sentiment on managerial compensation (Kuhnen and Niessen, 2012) as well as capital decisions in the form of acquisition abandonment (Liu and McConnell, 2013). Similarly, Peress (2014) studies the impact of news dissemination on financial market efficiency. Using newspaper strikes as information shocks, the authors find strike days are not only accompanied by significant drops in trading volume, but also lower dispersion and intraday volatility of stock returns. Dougal et al. (2012) show that not only the sentiment in financial news matters, as shown by Tetlock (2007), but journalists themselves possesses a causal influence on asset price returns.⁴ We extend this literature, highlighting the complementary role of media and civil society.

Our results show how civil society and media require one another to promote accountability. Accordingly, we contribute to rich empirical literature on the role of media and information in promoting political accountability. Ferraz and Finan (2008) and Larreguy et al. (2020) highlight how information on malfeasance through local media and audits, respectively, disciplines local politicians. Seminal work (Snyder and Strömberg, 2010) shows how media coverage interacts with electoral institutions to discipline congressional behavior in the US.⁵ In China, Qin et al. (2017) discuss the tacit approval of central authorities of controversial media as means of overseeing local-level corruption. Our study builds on this work and emphasising the potential use of media not only in

³This follows a large literature on political connections, including Fisman (2001)'s work on the political connections to General Soeharto in Indonesia.

⁴See for instance also Fang and Peress (2009), Bushee et al. (2010), Griffin et al. (2011), and Birz and Lott Jr (2011) on the media influence on financial markets as an information intermediary.

⁵In seminal work on the impact of television and radio exposure in Indonesia, Olken (2009) shows the importance of media in the governance of local road projects. While Olken (2009) reports negative effects on the attendance of village meetings responsible for planning and monitoring road construction, he finds no impact of television exposure on corruption associated with these projects.

constraining the behavior of conventional political actors, but also those of large multinational actors.

Our study highlights the role of international media and human rights advocacy in the developing world. We contribute to a literature on market reactions to firm misconduct, which has largely focused on high-income economies. This work has shown evidence that investors react negatively to convictions from corporate wrongdoing, such as fraud (*e.g.* Miller, 2006; Dyck et al., 2010), and new law constraining unethical behaviour by companies (Cousins et al., 2020).⁶ Our results parallel these findings. Recent papers by Krüger (2015), Capelle-Blancard and Petit (2019) and Cui and Docherty (2020) also show that stock markets react negatively to negative news about environmental, social and governance (ESG) events. Although much of this literature has focused on self-governance and the response to malfeasance in high-income economies, our results share commonalities.⁷ While focused on discrimination, recent historic work Do et al. (2020) also highlights the importance of the media. The authors investigate market reactions to the “Dreyfus Affair.” They find the infamous 19th-century episode of French antisemitism led to short-term devaluation of companies with Jewish connections, but show that (later) media revelations surrounding the scandal led to the rehabilitation of the French officer—and excess returns for these companies.⁸

This paper is one of the first, large empirical studies to show how the human rights spotlight impacts multinational asset prices. In our case, mining companies operating connected to the assassinations of civil society activists. Existing legal frameworks are often inadequate. It is well-recognised that formal laws—domestic and international—are inadequate to hold companies accountable. Though preliminary, our findings hint that the publicity strategies of human rights by revealing information to international markets. Even where formal justice is rare, these strategies may nevertheless have impact.

The remainder of the paper is organised as follows. Section 2 introduces our data, section 3 describes our empirical methodology and presents the results; section 4 presents the analysis of the media channel, section 5 provides a tentative explanation for the occurrence of the assassinations; and section 6 concludes.

2 Data Construction

We use event study methodology to study the impact of publicity surrounding salient, well-reported human rights violations: assassinations. To do so we collect 20 years of data on the assassination of activists connected to mining activity. The following sections describe this data collection effort. We first describe the definition and scope of the assassination events used in our analysis. Second, we describe the collection and basic contours of our dataset. Third, we describe the coding of companies. We conclude by describing additional financial and GIS data.

⁶Dai et al. (2015) show that media reports on past insider trading curb the profits of future insider transactions.

⁷*E.g.* through fines, lengthy court cases, or the withdrawal of a business license

⁸In a related work on discrimination, Huber et al. (2019) examine how the dismissal of Jewish managers in Nazi Germany led to a decline in firm stock market value.

2.1 Assassination Events and Descriptive Statistics

Assassinations and the Mining Sector - Our focus on assassinations is intentional. Since the early 2000s, the international human rights community has drawn attention to a rising trend in violence toward environmental activists (Butt et al., 2019; Hale, 2020). Specifically, the killing of activists connected to natural resource activity. This time of campaign is not idiosyncratic. Such “informational politics” of international human rights tend to coalesce around on the names of victims and the naming of targeted (states and firms) associated with these events (Keck and Sikkink, 1998; MCENTIRE et al., 2015).

Figure 2 shows the global trend in activist killings since 2008. Data come from our database on global activist assassinations. These events—including the victims and associated actors—embody the protagonists in international human rights campaigns.

By *assassinations*, we mean the intentional killing of prominent members of civil society.⁹ Since these individuals are locally prominent—as advocates or key players in their communities—we refer to their slayings as assassinations. These people include indigenous and tribal leaders; environmental and labour activists; members of the clergy; and more. Throughout the paper we use the terms assassination and extra-judicial killing interchangeably.

By construction, the assassinations we study are well-publicized. That is, those that draw human rights and media attention. We follow the human rights scholarship on “naming-and-shaming” campaigns, and focus on international publicity of these events (Ramos et al., 2007; Hafner-Burton, 2008; Peksen et al., 2014).¹⁰ We are interested in human rights reporting and publicity that is likely to reach international markets. Since we study the impact of this publicity, we do not consider killings that are unpublicised.

Our focus is on assassinations surrounding mining and mineral extraction activity. We do so for three reasons. First, this sector is one of the most deadly sectors for activists (Butt et al., 2019).¹¹ Second, it is a capital-intensive sector and one where equity financing, specifically, is common. As such, it is a sector where we are able to connect publicly traded firms to human rights events. Third, we limit our attention to a homogeneous, upstream production. By doing so, we attempt to limit the extent to which these products may face consumer boycotts in response to human rights reporting.

Event Data and Descriptive Statistics - Our assassination dataset covers 354 killings over 20 years. Our data collection process can, broadly, be summarised in the following way. First, we consider killings that are *publicly* reported in media or human rights campaigns. Second, we consider events where reporting connects a victim (or victims) to local mining and mineral extraction activity. Third, we then code the location where the death occurred. Fourth, we code the mining companies or projects named (if any) in relation to the event. Below we describe our data collection and coding, and provide details in Appendix B.1.

⁹This definition is not idiosyncratic. The operating definition in this paper and dataset tracks The Associated Press’ style guide, as well as that of the US National Public Radio standard. Academically, our definition aligns with the definition used in human rights scholarship (for example, DeMeritt, 2012).

¹⁰Scholarship in this area emphasized both individual dissemination of information from the NGOs themselves, the media reporting, and the UN.

¹¹See also: <https://www.globalwitness.org/en/press-releases/spotlight-criminalisation-land-and-environmental-defenders/>

Our assassination sample is collected using both algorithmic and human searches of international full-text media archives. These include full-text collections of International Herald Tribune; the Associated Press wire archive; popular news APIs (e.g. Guardian); and global news databases (e.g. LexisNexis).¹² Coding is done by research assistants and cross-validated by principal investigators.

Table 1 describes dimensions of our assassination data. With our first observation in 1998, the data set covers 496 victims of violence across 31 countries. Peru and Philippines are the most dangerous countries for mining activists. Figure 4 maps the geographic distribution of our event data assassinations.

2.2 “Associated” Companies

Table 1 shows publicly traded firms matched to—“associated with”—at least one event. Throughout this study, *association* means that a company, or their mining project, is named in media or HR reports surrounding an assassination event.

We take no stance on relationship between a firm and an event, and default to the information report in sources. Thus, an “associated” company does not mean a company plays an active role in organising or participating in violence (as shown below, it may be unlikely). Rather, a firm is tied to the event insofar as it—or its operations—are mentioned in publicity of an event. This point is key, as a publicly traded company may be an indirect owner of a project where the violence occurs.

Figure 3 breaks down the global distribution of assassinations and those companies associated with assassinations. Coloured panels (left) correspond to an event country. Their height represents the total number of assassination events in that country. Grey panels (right) correspond to the headquarter country for publicly traded firms. Their height represents the number of events connected to firms from that country.

Figure 3 shows a significant feature of our data and context. A majority of assassinations in our data are matched to firms headquartered in Canada, the United Kingdom, and the United States. Though these are advanced liberal, democracies, suits against multinationals for human rights abuses are exceedingly rare.

We hand-match the “nearest” publicly traded firm associated with each assassination. Matching public firms is done in three steps.

First, we determine whether a company named in human rights reporting is publicly traded.¹³ For consistency, we consider the most direct, publicly traded companies—except when the international, corporate owner is specifically tied to the event in reporting.

Second, when a company named in human rights reporting is not publicly listed, we search for its parent company by hand. We examine whether a named company is a subsidiary or joint venture of publicly traded companies (when the event occurred). We then verify this information using official firm websites; corporate reports; SEC filings; public business registers; and more.

¹²These include multilingual searches. Some media databases provide translations of international news coverage as well, such as LexisNexis.

¹³Special case arises if another public mining company holds shares of the company at the time of the event: i.e. the named company is not the ultimate owner.

Third, human rights reporting may refer to a mining project instead of the company responsible for the project. Geographic or administrative names often demarcate individual mining plants or project, as opposed to firms. For example, reports may refer to Rapu-Rapu Polymetallic Project (Philippines), as opposed to the company, Lafayette Mining. In these cases, we attribute ownership of the project to the publicly traded company using process described in step 2.

Figure 5 illustrates our coding process. The figure shows an excerpt from a *Guardian* newspaper report for an assassination in our sample.¹⁴ The piece identifies a victim (green), Ecuadorian indigenous leader José Isidro Tendetza Antún, and establishes they were in opposition to the mining activity (purple). Furthermore, it describes the victim’s violent death (yellow).

The example report describes a specific mining project, not the company running the project. The death in Figure 5 involved “Mirador copper and gold mine (blue)”, owned by *Corriente Resources Inc.* through a subsidiary *Ecuacorriente S.A.*. We code the ultimate owner using this information and public records. In this case, public corporate data states *Corriente Resources Inc.* was acquired in 2010. At the time of the event, it was owned by two publicly traded companies : *China Railway Construction Corporation* and *TongLing Nonferrous Metals Group Holdings*.¹⁵ Both publicly listed companies are coded as being “associated” with the event in our dataset.

When a particular project or company is not named in human rights reporting this assassination is not tied to a company. For example, consider Figure 5. If only the purple, not blue, highlighted information were available, the event would not be tied to a (publicly traded) firm.¹⁶

Furthermore, Figure 5 shows an important issue about event dates. Although, the date of the crime can be pinpointed as 28 November 2014 (pink), an event may make the news only after discovery of the body, etc. In this case, financial markets are likely to react only days after the *de facto* event date.

2.3 Further Data: Financial Outcomes, Geo-Location, and Control Companies

We collect daily stock return data for mining companies associated with event (see: Section 2.1 above), as well as returns for other companies operating within same country, during the year the event took place. Daily return data for 1998-2019, and additional firm-level data, come from the *Datastream* database.¹⁷

For mining projects in our data, location and company ownership data comes from the *SNL Minings & Metals* database, which we matched to our assassination data.¹⁸ This database also allows us to identify a robust set of control companies for each event-year because we can identify other mining companies with operations in the geographic vicinity of the event mine. Project ownership information is available at the annual level allowing us to track (treated and control mining) compa-

¹⁴The source article can be found here: <https://www.theguardian.com/world/2014/dec/06/ecuador-indigenous-leader-found-dead-lima-climate-talks>.

¹⁵See for instance <http://www.corriente.com/news/news.php> and https://www.banktrack.org/project/el_mirador_copper_mine for more information.

¹⁶Figure C.1 in the Appendix presents an example case in our dataset related to mining opposition but without a connection to a specific mining project or company.

¹⁷Market holidays are removed from the closing price timeseries. We use financial variables in a common USD denomination to account for currency fluctuations.

¹⁸See: <https://www.spglobal.com/marketintelligence/en/campaigns/metals-mining>. Other recent studies using the *SNL Minings & Metals* database comprise Berman et al. (2017) and Knutsen et al. (2017), who explore the impact of local mining operations on conflict, respectively corruption.

nies operating in an event-country over time. For example, if we observe an assassination event in Colombia in 2013, the control company set for this event comprises all publicly traded companies that own mining projects in Colombia in 2013 but are not directly associated with the assassination in media or NGO reports. Figure C.2 Panel A in the Appendix provides a graphical illustration of this example case.

Table 2 shows summary statistics for our financial data. For the construction of market returns, we rely on the MSCI (Morgan Stanley Capital International) stock indices. We match mining company securities to their associated MSCI country index based on where they are listed. If an assassination event falls on a non-trading day, the event date is assigned to the first trading day after the actual event date. Finally, we exclude thinly traded mining company securities from our sample.¹⁹ We apply this financial data to our event study, which we describe next (Section 3.1).

3 Event Studies: Activist Assassinations and Stock Returns

How does human rights coverage impact firms caught in the spotlight? We use event study methodology to estimate how—and if—the average stock price of these companies change following high profile assassination. We do so by examining the cumulative abnormal returns (CARs) of impacted companies. We estimate and track the returns of mining firms, starting on the date of assassination and estimating 1-day to 10-day returns starting from the event date.

We use two related strategies to study global market response to publicity around assassinations. First, we employ a traditional (“classic”) event study methodology. We use a traditional market model, and show that our results are robust to several parametric and non-parametric tests. Second, we consider the cumulative abnormal returns of affected (“treated”) versus control companies using cross-sectional OLS regressions.

In addition to examining the CARs of mining companies *after* publicity of assassination, we also consider returns *prior* to the date of violent events. We estimate the returns from the 10 to 1-days before the occurrence of an assassination, and test whether there are significant changes in the stock price prior to events occurring. These changes could indicate insider trading on prior knowledge of pre-meditated assassinations. More broadly, we are interested in whether firms associated with violence experience systematic changes in stock prices preceding violence.

3.1 The Classic Event Study: Estimation Strategy

Estimation and Event Timing - We first describe the classic event study methodology in the context of an assassination event timeline, shown in Figure 1.

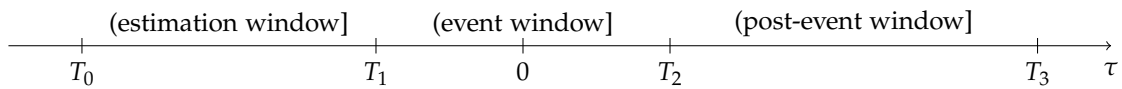


Figure 1: Event Study Time Line

¹⁹For completeness, we require that companies are traded at least 200 days out of the 250 trading days in the *estimation window*, which we turn to in the next section.

Abnormal returns are the difference between the observed return of company i and their expected return, absent the assassination event e . Abnormal returns AR around an assassination event can be written as

$$AR_{ie\tau} = R_{ie\tau} - E(R_{ie\tau}|X_\tau), \quad (1)$$

where X_τ is the information set that expectations about the “normal” returns are conditioned on, and τ denotes time relative to an event date.

Consider our timeline in Figure 1. The time around the assassination event at $\tau = 0$ can be split into three chunks, or windows. An *estimation*, *event*, and *post-event* window. Following the event study literature (MacKinlay, 1997), normal returns for company i are determined by estimating a market model over the *estimation window* ($\tau = T_0 + 1, \dots, T_1$). The market model’s linear specification for firm i affected by assassination e is

$$R_{ie\tau} = \alpha_{ie} + \beta_{ie}R_{ie\tau}^M + \epsilon_{ie\tau}, \quad (2)$$

where $R_{ie\tau}$ is the observed daily stock return for firm i , $R_{ie\tau}^M$ is the return of the market index where company i is listed, and ϵ is the residual.

For estimating the model parameters, we choose an *estimation window* of 250 trading days, ending 30 days *before* the event (day 0).²⁰ For this calculation, we require securities to be traded at least 200 out of the 250 trading days, and 8 out of the 11 trading days during the *event window* from $T_1 + 1$ to T_2 under our baseline specification.

Estimated abnormal returns, $\widehat{AR}_{ie\tau}$, and cumulative abnormal returns from τ_1 to τ_2 , $\widehat{CAR}_{ie}(\tau_1, \tau_2)$, during the *event window* are given by

$$\widehat{AR}_{ie\tau} = R_{ie\tau} - \hat{\alpha}_{ie} - \hat{\beta}_{ie}R_{ie\tau}^M \quad (3)$$

$$\widehat{CAR}_{ie}(\tau_1, \tau_2) = \sum_{\tau=\tau_1}^{\tau_2} \widehat{AR}_{ie\tau}, \quad (4)$$

where $T_1 < \tau_1 \leq \tau_2 \leq T_2$. The conditional variance of the respective cumulative abnormal return is

$$\sigma^2(\widehat{CAR}_{ie}(\tau_1, \tau_2)) = L_2 \sigma_{\epsilon_{ie}}^2 \left[1 + \frac{L_2}{L_1} + \frac{L_2 \left(\frac{\sum_{\tau=T_1+1}^{\tau_2} R_{ie\tau}^M}{L_2} - \mu_{ie}^M \right)^2}{L_1 \hat{\sigma}_{ie}^M} \right], \quad (5)$$

²⁰This follows Li and Lie (2006) and Acemoglu et al. (2016).

where $\sigma_{\epsilon_{ie}}^2$ is the variance of the regression residual $\epsilon_{ie\tau}$, and $L_2 = \tau_2 - \tau_1 + 1$ is the length of the “aggregation” period. The terms μ_{ie}^M and $\hat{\sigma}_{ie}^M$ are the mean and variance returns over the estimation period, $L_1 = T_1 - T_0$. The second component of $\sigma_2(\widehat{CAR}_{ie}(\tau_1, \tau_2))$ accounts for the additional variance introduced by the sampling error in $\hat{\alpha}_{ie}$ and $\hat{\beta}_{ie}$ (Salinger, 1992; MacKinlay, 1997).²¹

Inference in "Classic" Event Studies For completeness, our study uses of multiple test statistics for inference. We show that Kolari and Pynnönen (2011)’s GRANK test specifically speaks to our setting where the day of pricing is likely to deviate from the actual assassination date.

First and foremost, our study is about the *average* effect of publicity surrounding assassinations. To consider overall inference of events, we must aggregate cumulative abnormal returns across all company-event pairs N . Thus, the average CAR, and its variance are,

$$\overline{CAR}(\tau_1, \tau_2) = \frac{1}{N} \sum_{j=1}^N \widehat{CAR}_{ie}(\tau_1, \tau_2) \quad (6)$$

$$\sigma^2(\overline{CAR}(\tau_1, \tau_2)) = \frac{1}{N^2} \sum_{j=1}^N \sigma^2(\widehat{CAR}_{ie}(\tau_1, \tau_2)) \quad (7)$$

Assuming normally distributed security returns, absent clustering (overlapping event-windows), $\overline{CAR}(\tau_1, \tau_2)$ follows a normal distribution with mean zero and variance $\sigma^2(\overline{CAR}(\tau_1, \tau_2))$ (MacKinlay, 1997).²² We refer to this test as the “Normality” test.

There are many reasons researchers wish to relax “normality” assumptions.²³ We go beyond the standard (“Normality”) tests in three ways.

First, we utilise two additional parametric tests suggested by Boehmer et al. (1991) (“BMP”) and Kolari and Pynnönen (2010) (“Adjusted-BMP”). The first parametric test, BMP, scales abnormal returns, adjusting for differences in the variance of pre-event residuals (building on Patell (1976)). Intuitively, more volatile securities are down-weighted to prevent them from biasing the detected average event effect; and the BMP test goes beyond Patell (1976)’s test by accounting for changes in event-induced volatility.

Second, we use a refinement of BMP which accounts for event clustering. Though we suspect clustering issues should be minor in our setting, we nevertheless account for the potential bias from cross-correlation of abnormal returns. We do so by using Kolari and Pynnönen (2010)’s adjustment (“ADJ-BMP”).²⁴

Our third—and preferred—approach relaxes the assumption of normally distributed returns, and we believe is suited to the informational environment of our study. We implement the non-parametric generalized rank t -statistic (GRANK) from Kolari and Pynnönen (2011). Kolari and Pynnönen (2011) argue that their GRANK statistic performs well in situations much like our

²¹It follows from (5), that the variance of the abnormal returns during the *event window* is given by $\sigma^2(\widehat{AR}_{ie\tau}) = \sigma_{\epsilon_{ie}}^2 \left[1 + \frac{1}{L_1} + \frac{(R_{ie\tau}^M - \mu_{ie}^M)^2}{L_1 \sigma_{ie}^M} \right]$.

²²The normality assumption requires the absence of clustering in order to set the covariance terms in (7) to zero.

²³Following a long literature in event study methodology (Boehmer et al., 1991; MacKinlay, 1997).

²⁴In our study, *at most* four public firms are associated with any given event. The correction might, however, be warranted as our market model extracts *less* correlation from regression residuals than the alternative Fama-French model.

assassination setting. Mainly, they show their generalized rank statistic outperforms parametric (and non-parametric) tests in studies where 1) exact event day may be unknown and 2) long event windows are used. We feel this is particularly useful for us, as sensitive information in our setting may percolate slowly into the market and the precise day of the pricing—in contrast to the assassination date—is uncertain. Additionally, the GRANK statistic is robust to event-induced volatility, serial correlation and event-day clustering. For a further technical discussion of our test-statistics, see Appendix, section A.1.

For completeness we show our main event study results across test-statistics. We turn to these results now.

3.2 Classic Event Study: Results

Table 3 shows our core results. Overall, we find assassinations events lead to negative abnormal returns for firms associated with violence. Table 3 shows that negative effects start soon after the date of the assassination, and these effects grow through time. Both in a) magnitude and b) significance. Importantly, the effects are significant across tests.

The pattern in Table 3 is consistent with sensitive information gradually diffusing through the market. On the day of an assassination, we see little market reaction, followed the next day by a (borderline) significant effect of around -0.7 percentage points. This initial reaction is followed by a stable decline the next four days, and a steep—and robustly significant—decline from day 5 through day 10 after the event. This cascade suggests that market participants may first gather additional information surrounding an event, before pricing the expected reputational and legal costs for an “associated” mining company.

The average cumulative abnormal return is -2.0 percentage points 10 days following the event. These estimates are significant: at the 1 percent level using our preferred GRANK statistic (column 6), and at the 5 percent level with the “BMP” (4) and adjusted BMP statistics (5). Furthermore, by considering multiple test statistics, our findings in Table 3 suggest differences in volatility across securities could bias inference if not accounted for: comparing our adjusted test and non-parametric statistics to the “normal” test in column (3). Meanwhile, the clustering issues in our setting seem negligible—the differences between the BMP and adjusted BMP tests are small (columns 4 versus 5).

All our test-statistics account for non-trading days of securities, adjusting for the length of the *estimation* and *event window*. However, our results are robust to more conservative trading day criteria. Appendix Table D.1 shows results are unchanged when we require companies be traded each day within an event window, and 225 out of 250 days during the estimation window. This adjustment drops seven company-event pairs, leading to a slight decline in the magnitude of our CARs to -1.5 percentage points ten days after an event. As well, these results are still highly significant and robust across tests. The difference in point estimates may be driven by a trading halt for highly affected securities. On the other hand, the securities of small mining companies are also less frequently traded; we may expect these firms to be more vulnerable to disruption following an event. Thus, stricter requirements on trading frequency might disguise the true effect of assassination events and we default to our original cut-off criteria.

Next, we turn to the days leading to the event. By considering the days before an assassination, we test whether market participants traded on foreknowledge events. Furthermore, where events are planned, the “authorization” date of assassinations is unknown to us. A reasonable assumption is that—if private information exists—it should be priced close to the actual event date, when the likelihood of execution can be best assessed by insiders.

We test this “prior knowledge” hypothesis, and for pre-treatment movements in returns, by aggregating abnormal returns *backwards* starting on the day before the event (c. for instance also Dube et al., 2011). Appendix Table D.2 reports these results. The average abnormal return on the day before the event is positive, while the cumulative abnormal return over the ten days before an event is close to 0—slightly negative—and insignificant across test statistics. These results indicate two key findings. First, the market did not price prior knowledge of assassinations. Second, and importantly, our core event study results are not *merely* picking up a downward pre-trend in the asset prices for those companies associated with violence.

3.3 OLS Event Study Using Treatment and Control Firms

We now build on the traditional event study design above by using the full variation of our event data. Relative to many small-N event studies, our event data (over 160 companies-event pairs) introduces rich panel and cross-sectional variation to further explore market reactions to assassinations. This section introduces an intuitive OLS estimation strategy that makes full use of this data.

Our regression framework examines the relationship between the publicity around assassinations and stock returns of companies differentially exposed to violence. Intuitively, we compare the cumulative abnormal returns for companies whose projects are directly associated (named) with these events versus companies operating in the same sector, country, and period of the assassination. The regression model we consider can be written as

$$CAR_{ie}(\tau_1, \tau_2) = \alpha + \delta D_{ie} + \mathbf{X}'_{ie}\phi + \gamma_e + \epsilon_{ie}, \quad (8)$$

where $CAR_{ie}(\tau_1, \tau_2)$ is the cumulative abnormal returns for company i . For each event e , we estimate CARs between for a window between periods τ_1 and τ_2 . The indicator D_{ie} denotes treatment, and is equal to one if a company is associated with an event and zero otherwise.

The coefficient of interest δ captures the average difference in CARs between firms treated versus control firms—companies with operations in the same country, sector, over the event period. Our empirical strategy is valid if, absent violence during the event period, we would not observe systematic differences in the returns of treated (“associated”) versus control firms.

Our preferred specification (8) includes event-specific effects, γ_e . Including event-specific effects controls for common market reactions around dramatic events, such as shifts in market sentiment toward the event country, or increased excess volatility.

Alongside (8), we present a specification which includes company fixed effects (γ_i). This specification uses the dimensions of our data. Depending if a company was *merely* active in the country

during an event-period or directly tied to an assassination it can be part of both, the control and treatment group, for different assassination events.

The baseline specifications includes a set of time-variant, firm-level controls, X_{ie} . These are firm size, total assets, and leverage (total debt to capital). Small or highly-leveraged firms may be more dependent on specific mining projects, and thus differentially impacted by events in our study. Disruptions of projects may be more punitive for smaller (more leveraged) firms. To address issues of “bad controls,” we use lagged values for controls. Standard errors are robust to heteroscedasticity and clustered at the event level.

Our set of control companies has a number of advantages. First, we account for exposure to political risk events common to mining companies operating in a given location at the time of the event. This allows us to account, among other things, for incidents where violence against activists changes the national sentiment against the mining industry. Second, we follow the rationale established by Guidolin and La Ferrara (2007); we wish to compare treated companies to those with a similar “comparative advantage” for operating in political risk environments (p.1987). Last, by limiting the control group geographically, we help control for commodity price fluctuations similarly impacting mining companies operating in similar commodity markets and the same domestic market.²⁵

3.3.1 OLS Event Study Results

Figure C.3 reports the core results. The top panel presents regression coefficients ($\hat{\delta}$) for our baseline specification with event fixed effects, the bottom panel presents estimates for company fixed effects. In total, 42 *individual* regression estimates are displayed with the vertical axis showing the (τ) days after (before) the event. We present 95 percent confidence error bands.

Figure C.3 shows a clear pattern. In the days preceding and event, CARs are not significantly different between exposed and non-exposed firms. Returns for associated firms are, if anything, slightly positive.

Importantly, a consistent negative decline in abnormal returns for treated versus control companies emerges, following an assassination. We see no market reaction on the event day. Soon after, markets start responding to the assassination reporting. Two days following an assassination, CARs amount to -1.0, respectively -1.4 percentage points and are statistically significant. Thereafter, CARs remain negative, gradually dropping five days after the event. By day ten, the abnormal returns for exposed mining companies are between -2.2 to -2.4 percentage points. These findings reinforce our interpretation of our classic event study results. Financial markets take time to absorb the publicity and assess risk for mining companies tied to the assassination event. In fact, the pattern in C.3 corresponds to that found in our traditional event study results.

Results are particularly strong for our specification using firm fixed effects (Figure C.3, row 2). When we use only within-firm variation and a company implicitly serves as its own “control group”, the significant difference in CARs provides strong evidence for the presumption that the actual or expected disclosure of a company’s role in the assassination event is at the root of the estimated effect, a mechanism we will investigate further in Section 4.

²⁵For example, commodities mined in Columbia from 2013-2017 comprise coal, nickel, gold, emerald and iron according to EITI records, with about 70% of the companies active during those years mining coal.

Figure C.3 in the Appendix shows that our baseline results are robust to accounting for potential non-linear effects of covariates by including cubics in size, leverage and additionally profitability (return on equity) (column 2).²⁶ Further, the detected pattern remains qualitatively and quantitatively unchanged for specifications with a less conservative set of fixed-effects—i.e. year (γ_y) and headquarter country (γ_h) effects (rows 2 and 3).²⁷

We quantify these results in Figure C.4 and show they are economically meaningful. Figure C.4 shows the estimated loss in market capitalization for the median treated company. Results are shown for each day. Dots correspond to baseline estimates. Error bars corresponds to the minimum and maximum loss across specifications. Figure C.4 shows the median treated company is estimated to loose between 100 and 150 million USD in market capitalization over the ten days following the event.²⁸

3.3.2 Additional Robustness Checks for OLS Results

We show the OLS event study results above are not driven by a particular, individual nation or company. Nor are they sensitive to specific events or window size. The following section presents robustness results by considering robustness of our OLS results across these dimensions.

Figure 8 presents a “leave-one-out” analysis of our OLS results, and shows that estimates are not driven single countries or firms. Our baseline regression results are shown in bold, and permutations are shown in light grey.

In Figure 8 Panel A, we re-estimate our baseline specification, but sequentially drop individual event-countries from our sample. Visually, our the negative effects are similar across sample permutations. Panel A shows a clear and gradual decrease in abnormal returns over the 10 days following the event. As well, abnormal returns in the days leading up to the event remain slightly positive, though close to zero. Importantly, the core pattern of our baseline results are robust to excluding the two deadliest countries for mining activists: Peru and the Philippines (shown in dashed red).

Figure 8 Panel B repeats the same exercise, but for sequentially dropping firms. Here too our baseline results are qualitatively similar after sequentially dropping firms. The homogeneity in across these light grey bands indicate that our findings are not driven by particular “bad actors.” This suggests are results are applicable to publicly traded mining multinationals, broadly.

Our OLS results are robust across different types of assassinations and similar events. Table D.4 shows that post-event assassination estimates are qualitatively unchanged when we exclude unsuccessful assassinations (columns 1 to 5)—or, assassination attempts. Similarly, when we exclude activist killings during protests (columns 6 to 10). Although the point estimates here follow a similar pattern to our baseline results, we exclude a substantial number of events. Excluding 42 observations, as in the previous case, reduces the power and significance of the results.

²⁶C. for instance Acemoglu et al. (2016).

²⁷In total, Figure C.3 presents 210 *individual* regression estimates. Columns correspond to two types firm-level controls used across different specifications. Each row shows five types of fixed effects. Each panel (cell), thus, shows a set of estimates for one of ten specifications. Corresponding regression tables are reported in Appendix Table D.3.

²⁸Figure C.4 in the Appendix presents the distribution of losses across companies for the *baseline* specification. Demonstrating large, and economically meaningful losses for companies associated—even loosely—with violence. For clarity, due to long-tails, we do not display losses above the 90th percentile.

Our baseline results are not driven by outliers. Table D.4 shows dropping CARs larger than the 99th percentile or smaller than the 1st percentile (columns 11 to 15) does not alter the substance of our results. The assassination coefficients are similar in magnitude. In fact, they are more precisely estimated across specifications, with the exception of company fixed effects. This latter case may indicate that the association with an assassination event constitutes an extreme observation for several companies that have been tied to an event at least once. Broadly, this supports the idea that assassinations can have substantial effects on single firms.

Finally, we expand the event window to twenty days prior and post the event day to investigate the possibility of pre-trends more closely and test for a potential reversal in the estimated effect after day 10.

Appendix Figure C.5 plots the assassination coefficient estimates and the 95 percent confidence intervals for our baseline specification from over 40 specifications.²⁹ After the event date, we observe a monotonic decrease in returns until day 13 that seems to be permanent. Before the event date, estimates are positive and insignificant. We conclude that the effect of publicised assassination events is persistent and cannot be explained by pre-existing trends in the stock price.³⁰

4 Mechanisms

This section sheds light on the underlying mechanisms that explain how civil society campaigns around activist assassinations have such an immediate and relatively large impact on the stock price of associated companies. The overall process, can be summarised as follows: Civil society groups' main strategy is to name the mining companies linked with the event and use the global news media to make the general public and the financial markets aware of the human rights abuses. The likelihood that a story about an activist assassination gets wider attention and reaches financial decision makers, depends on the amount of other newsworthy events on the days around the assassination. Investors whose strategy is more sensitive to (negative) news will relatively quickly decrease their holdings in associated stocks to avoid short-term losses. This sudden reaction by institutional investors who follow an event based strategy, leads to the short-term drop in share prices depicted in our main results.

4.1 Vicinity vs. Associations to Assassinations

So far, our classic event study, and the regression results, present a consistent story: assassination events significantly—and negatively—impact the returns for companies associated with violence. In this section we further unpack what may be driving these results. Specifically, we ask whether our results are driven by disruptions surrounding the violence *per se*? Rather than the market pricing the impact of being named in connection with the killing of an activist, our results may be driven by other, physical disruptions surrounding the event or unobserved regional violence correlated with the assassination event.

²⁹Note that we adjust the minimum trading days requirement for this analysis: for companies to be considered, they have to be traded on at least 15 of the 21 days post the event, respectively 15 of the 20 days prior to the event.

³⁰Figure C.6 in the Appendix provides further support for the persistence of the publicity effect. Graphing the average CARs for *associated* companies in the 90 days following the event reveals that there is no reversal in the estimated effect even when considering very long-time horizons. Note that C.6 simply displays average CARs when underlying the market model (see Section 3.1), not OLS estimates.

In this section we address this concern by comparing the asset price responses of firms publicly exposed to the event to other companies operating within the same administrative unit in the event country. We use the regression framework from above to compare the CARs of treated firms to untreated firms operating in the same sector and same administrative unit as the treated firm.

For this robustness check, we construct a new set of control companies. We do so by matching the geolocation of our assassination events to properties in the *SNL Minings & Metals* database. Specifically, we connect assassination events to mining projects in the same *Admin1* region (see Figure C.2 Panel B in the Appendix for a graphical illustration).³¹ For 92 events in our sample, we were able to match at least one publicly traded control company operating in the same region as a company “exposed” to an violent event. Figure 9 shows the results of our analysis. The post-event assassination coefficient estimates for our Admin1 subsample are presented in Panel B, and Panel A displays the estimates for our baseline sample.

While we lose observations through our geo-matching process, we nonetheless observe a gradual relative decrease in the CARs for treated firms. CARs are -1.4 percentage points lower, ten days after an event for our baseline specification. Appendix Table D.5 shows that our estimates are broadly negative across specifications and of similar magnitude. With the exception of column 9. Column 1 shows the specifications where we use maximal variation of our data, excluding fixed effects. Here, the average CAR is -2.8 percentage points, significant at the 5 percent level. Our results indicate that the negative impact we observe are likely not driven by the proximity to violence and spatial disruptions from these events. This reinforces that our results are likely driven by firms being associated to violence via the media channel. Furthermore, our results suggest that traders may be well informed about circumstances surrounding events, which we investigate further in Panel C.

In Figure 9 Panel C, we use publicised assassinations events of activists where no company was linked in the media reports. In other words, we consider assassinations where no firm has been publicly named in connection with the violence.³²

With this in mind, we rerun our regression analysis. However, we consider slightly different treatment and control groups for these “unattributed events.” As we did before, we geolocate the “unattributed events” in our sample. Using this location information, we then consider *all* companies within the same *Admin1* region (for most countries, roughly the provincial level) of the event as treated ($D = 1$). All other companies in the event-country are in the control group ($D = 0$).

The results show that market participants appear to react with a time lag.³³ This behaviour is consistent with a lengthier information gathering process for events that, *ex post*, have no direct company tie. Furthermore, these results suggest that financial markets might have access to information that is not publicly available.

³¹We do so through ArcPy/ArcGIS. Given the geographic resolution of assassination events and projects in the database, there will be uncertainty in the precision of our “matches” within an Admin1 region. However, even with this uncertainty, we suspect these matches are nevertheless informative. Opposition to mining usually arises due to a local land conflict—such as conflict over indigenous property rights—and these conflicts are geo-spatially correlated.

³²Note that we cannot rule out that reports exist that establish a link between the assassination and a mining company or project. However, we are confident that the information is—at least—not easily accessible given the extensive time that has been attributed to researching the events.

³³The estimated coefficients across specifications are presented in Table D.6 in the Appendix.

4.2 The Role of the Media

We now turn our attention on the nexus between publicity of activist assassinations and the media. The role of the media in this context is twofold: First, journalists are often among the first ones to shed light on activists assassinations and media outlets help to communicate these stories to a broader audience. In most cases, media outlets are also the main source of information for investors about these type of market relevant events. Second, increased media attention for an assassination event, means more negative publicity and a higher chance of public backlash against the resource company tied to the assassination. As such, media attention for an event can be one of the main mechanism driving our baseline results.

A major empirical challenge analysing the media channel is that proxies for media attention are likely to be endogenous. For example, simply counting the number of news reports about a particular assassination might be a good measure for media attention, but at the same time could be driven by negative stock market reactions to the assassination event.

Instead of relying on a direct, but endogenous, measure for media attention for assassination events, we use an exogenous variable driving the likelihood that an assassination receives media attention. In particular, we use the daily news pressure index of Eisensee and Strömberg (2007).³⁴ We expect the likelihood of the event to get reported or gain attention to be smaller if the event coincides with a “high news pressure day”—defined as an above median news pressure day for the period from 1998 to 2018.³⁵ Even if investors obtain information from private sources and not the media, news pressure should serve as an indicator for the information demand of financial markets on the event day. For instance, news about natural disasters or recession forecasts are likely to both, dominate trading behavior and feature in the top news segments. In contrast, assassinations of mining activists are highly unlikely to be broadcasted in the top three news segments, making daily news pressure an arguably exogenous measure in our setting.

To estimate the heterogeneity of the assassination effect by the likelihood of media attention, we expand our baseline regression model by an interaction term between the treatment indicator D_{ie} and a dummy variable N_e equaling one if the event falls on a high news pressure day, and 0 otherwise:

$$CAR_{ie}(\tau_1, \tau_2) = \alpha + \alpha^N N_e + \delta D_{ie} + \delta^N D_{ie} N_e + \mathbf{X}'_{ie} \phi + \gamma_e + \epsilon_{ie}. \quad (9)$$

Note that we allow for differential intercepts of high and low news-pressure days to account for generic differences in trading behavior on high news pressure days.

Figure 10 shows the influence of news pressure on market reactions graphically for all three samples. The red line in the top panel corresponds to the cumulative abnormal return if the event day coincides with a high news pressure day ($\delta + \delta^N$), whereas the black line captures the effect of being tied to an event if the assassination date falls on a below median news pressure day (δ). The bottom panel graphs the difference in estimated effects (δ^N). Each panel also reports 95% confidence intervals.

³⁴Eisensee and Strömberg (2007) define *daily news pressure* as the median number of minutes a news broadcast devotes to the top three news segments in a day. For more details on the construction of the daily news pressure see Eisensee and Strömberg (2007) Section II.C and Appendix V.B.

³⁵Recall that 1998 constitutes the the first year in our assassination dataset and 2018 is the last year for which the news pressure is available.

For the baseline sample, the results show a significant and continuous decline in abnormal returns for associated companies if the assassination event falls on a low news pressure day, whereas the coefficient is indistinguishable from zero if the event coincides with a high news pressure day. The bottom panel plots the gradual divergence in cumulative abnormal returns. By day ten, the difference in cumulative abnormal returns is 3.9 percentage points and significant at the 5% level. An attenuated but similar divergence in effects is visible for the Admin 1 control subsample in column 2, while a noisily estimated inverse pattern emerges for the sample without explicit company ties (column 3). The latter, however, should be interpreted with caution, as the absence of direct company associations suggests an inferior role of the media in the dissemination of information.

A potential concern with this identification strategy is that the integration of media markets over time increased the availability of breaking news stories. Figure C.7 in the Appendix shows that the results are virtually unchanged if we detrend the *daily news pressure* before applying the median split.³⁶ Moreover, the estimates are qualitatively similar if high news pressure days are defined as a daily news pressure above the 75th percentile or if we consider the day after the event (Figure C.8 in the Appendix). In the latter case, we still find a visible but attenuated difference in effects, which is consistent with the notion that breaking news stories dominate the news for more than a day.

Overall, we believe one plausible interpretation of the results is that financial markets price in the company's continuous risks and costs of the initial public exposure of the company's association with human rights abuses.

Next, we turn to a related question. Can more transparency in the event country's mining industry support human rights organisations—in cooperation with media outlets—in their mission to hold corporations accountable for misconduct?

We address this question by interacting our assassination indicator with a dummy variable for event country membership in the Extractive Industries Transparency Initiative (EITI) at the time of the event. The initiative commits member countries to fully disclose taxes and payments made by mining companies to the government and serves as an indicator for quality of governance (Fukuyama, 2016).³⁷ For instance, Berman et al. (2017) find that EITI membership reduces the likelihood of mining related, localised conflicts in Africa. The results in Appendix Figure C.9 provide further support that transparency can amplify the publicity effect on associated multinationals' stock value. Published assassination events that occurred in a country which was an EITI member at the time of the event have a relatively stronger, negative effect on the associated mining company's market value compared to events that happened in non-EITI member countries.

4.3 Who reacts to publicity?

Our findings in the previous sections show that investors react negatively to assassination events. In the next step, we investigate if institutional investors—sophisticated, informationally-sensitive “big players” (e.g. Puckett and Yan, 2011; Hendershott et al., 2015)—respond to reports of human right violations for companies in their portfolio. For instance, Mccahery et al. (2016) report that socially “irresponsible” corporate behavior is considered a very important trigger for shareholder

³⁶Eisensee and Strömberg (2007) note a slight upward trend in the daily news pressure for the 1968-2003 period.

³⁷Data on “join” and “leave” dates of member countries is retrieved from the EITI API version v2. For more details on EITI, see Section 5 and <https://eiti.org/>.

activism by 72% of the surveyed 143 institutional investors.³⁸ Large scale sovereign wealth and pension funds have been early adopters of ethical investment policies and are increasingly divesting from stocks that do not meet certain ESG criteria.³⁹ Therefore, certain types of institutional investors could be more responsive to activist assassinations and sell off their holdings in the companies tied to the event.

We examine the relation between news about company “ties” to assassination events and institutional ownership using the specification:

$$IO_{it} = \alpha + \delta D_{it} + \mathbf{X}_{it}'\phi + \gamma_i + \lambda_{1i}t + \lambda_{2i}t^2 + \epsilon_{it}, \quad (10)$$

where IO_{it} is the ratio (as percent of market capitalization) held by institutional owners for quarter t of company i , D_{it} a dummy variable equaling one if company i was associated with at least one assassination event e in quarter t , and a set of annual firm characteristics $\mathbf{X}_{it}$ —i.e. size and leverage lagged by one year. Additionally, we control for company fixed effects (γ_i), as well as company-specific quadratic time-trends ($\lambda_{1i}t + \lambda_{2i}t^2$).

We obtain data on institutional ownership from the Factset Ownership database. Factset has been widely used in empirical finance (Aggarwal et al., 2011; Dyck et al., 2019) and reports institutional investors’ equity holdings collected globally from fund reports, regulatory authorities (e.g. 13F reports in the United States), fund associations, and fund management companies.⁴⁰ Importantly, Factset data spans from 2000 to 2017, at quarterly frequency, and covers 83 out of the 87 traded mining companies associated with assassination events in our sample. Or 153 event-company pairs. The average probability to observe an event in a quarter is 3.6 percent.

Table 4 Column 1 shows institutional ownership, on average, does not significantly change in response to assassination events. A possible explanation is that institutional investors with holdings in mining companies follow a more long-term strategy and are less responsive to short-term events.

Across institutional investors, however, there are likely different objects and strategies. Table 4 (columns 2-8) shows the results when re-estimating the regression for each type of institutional investor. Here, hedge funds significantly decrease their average holding position by about 16.5 percent,⁴¹ while we observe no significant reactions from other institutional investors.⁴²

The hedge fund results in Table 4 are dovetail with the literature on hedge fund strategy. They are consistent with the notion that the hedge funds’ shorter investment horizon (Cella et al., 2013) inclines them to monitor corporate behaviour closely and respond rapidly to costly information disclosure (Gargano et al., 2017), in particular, following revelatory news events (Huang et al., 2020).

In contrast, institutional owners that invest heavily in mining companies appear to prioritize long-term financial goals and might first resort to shareholder activism rather than liquidation of their

³⁸Other recent studies (e.g. Dyck et al., 2019; Chen et al., 2020) have also found evidence for a positive relationship between institutional ownership and corporate social responsibility scores.

³⁹See for example “Norway prepares to dump up to \$3.7b in Aussie shares”, Jun 13 2019, Australian Financial Review, or “Norwegian wealth fund blacklists G4S shares over human rights concerns”, Nov 14 2019, The Guardian.

⁴⁰In particular, we rely on “Institutional Ownership Summary Statistics by Firm” as developed by Ferreira and Matos (2008) and provided by WRDS. Data on annual firm characteristics are obtained from the *Factset Fundamentals database*.

⁴¹Note that $\frac{0.0034}{0.0206} \times 100 \approx 16.5\%$.

⁴²Our results are unchanged when we restrict the analysis to the mining companies in the baseline event study sample (Table D.7).

position.⁴³ Their inertia is highlighted by the positive—though imprecisely estimated—coefficient for the top five institutional owners. On average, these five make up about 42.6 percent of total institutional ownership.

We do not interpret our results as certain institutional investors acting out of moral virtue.⁴⁴ Rather, the human rights spotlight may prompt sophisticated investors into action, in anticipation of other market participants. In a counterfactual setting, without scrutiny from human rights reporting, the “noiseless” elimination of mining opposition may not inspire the same response.

The downward trend in prices following events seems compatible with the behavior of institutional investors relative to retail investors. Compared to institutional counterparts, retail traders may receive and respond to news slower. This is consistent with the gradual dip in prices, after an event. Similarly, market reactions are attenuated on days dominated by other newsworthy events. On such days, news is less likely to reach retail investors versus news-sensitive institutional traders. Together with Table 4, our results suggest that skilled investors, such as hedge funds, are able to predict responses of retail investors.

5 The Limits of Publicity - Tax Revenues and Assassinations

Given the losses from assassinations, why do publicly traded mining companies continue to be associated with these events? Multinationals are sprawling and complex. Their upstream operations may be operated by actors whose incentives differ from their owners.

Local parties benefiting from mining projects may have an incentive to suppress or eliminate opposition, and may not bear the risk of these actions. Domestic government and state-aligned paramilitaries are examples of such agents, where uninterrupted production will result in higher royalties and tax revenues. The relative gains of engaging in malfeasance can be higher for mining operations who earn significant revenue from mining projects. If so, the likelihood of observing an assassination should be increasing in the share of taxes paid to a host-country.

We explore this by constructing new data on the local public finance of mining companies. We do so by collecting reports published by the Extractive Industries Transparency Initiative (EITI)—itself an international civil society organisation. Using these reports, we construct annual measures of tax shares paid to host governments by mining firms.

As members of the EITI, nations commit to disclosing payments from local companies to the government; these revenue streams ordinarily cover payments from subsidiaries and joint ventures. We consequently must determine the ownership structures to match EITI records with our assassination event dataset.

⁴³ An illustrative example is the Church of England’s sale of its Exxon Mobil Corp. shares in October 2020, after a failed attempt to push through a resolution to split the chief executive and chairman roles at the shareholders meeting in May earlier in the year, in response to the company’s failure to provide an adequate response to the climate crisis. For details, see <https://www.bloomberg.com/news/articles/2020-10-08/church-of-england-pensions-board-has-divested-from-exxonmobil>.

⁴⁴ For instance, Och-Ziff Capital Management, a large, publicly traded hedge fund, invested 100 million USD in Camec, which used the money to purchase a joint venture with the state-owned Zimbabwe Mining Development Corp. (ZMDC). The deal funneled about 100 million USD as a cash loan to Mugabe’s government in need of funds at the time. For more details, see: <https://www.bloomberg.com/news/articles/2014-08-21/mugabes-bailout-och-ziff-investment-linked-to-zimbabwe-despot>.

Specifically, we hand-code ownership shares using information published in annual reports of publicly traded companies and—if not available—we rely on Bureau van Dijk’s *Orbis* database. As a convention, private companies are coded as their own owner, i.e. we do not account for ownership shares of private individuals. Tax revenues of subsidiaries and joint ventures are distributed to owners in accordance with their shares.⁴⁵ For instance, in 2014, *Anglo American* owned 81.90 percent of the Peruvian mining company *Anglo American Quellaveco S.A.* and *Mitsubishi* owned 18.10 percent. Consequently, 202,232 USD of the 246,925 USD in taxes and royalties to the Peruvian government are attributed to *Anglo American* while the remainder is attributed to *Mitsubishi*. For each country-year pair (report), revenues are subsequently aggregated at the owner-level and divided by the total amount of tax revenues from the mining industry to obtain the *tax share*. Summary statistics on the tax shares, disaggregated by event-country are presented in Table 5.

One caveat of EITI records is that they only cover a limited set of countries and years in our assassination dataset. We code ownership shares for all years available in the EITI database for countries which have experienced at least one assassination event in the past. It is worth noting, that we are not limited to public companies in this analysis. Thus we are able to additionally match private companies to assassinations in our data. For completeness, we retain potentially interesting cross-country variation and do not exclude countries if an assassination event falls outside of EITI coverage period.

We use a linear probability model to estimate the relationship between a company’s tax share and the likelihood to observe an assassination event. Specifically, we estimate the following

$$\text{assassination}_{icy} = \beta_1 T_{icy} + \gamma_{cy} + \epsilon_{icy}, \quad (11)$$

where $\text{assassination}_{icy}$ is a dummy variable that takes the value one if an assassination event in country c in year y is associated with company i and T_{icy} corresponds to the tax share of company i in country c in year y . In our baseline specification we include country $\times$ year effects (γ_{cy}), which account for time-varying economic and political developments in countries. Thus, we focus on variation within country-year, which may alleviate concerns about sample selection.

Table 6 shows a positive relationship between the tax share and the probability of observing an assassination event. Column 1 presents the significant and positive unconditional correlation coefficient of 13.8 percentage points. The magnitude of the coefficient increases to 17.4 percentage points after we account for structural differences in the mining industry and average prevalence of assassinations with country fixed effects (column 2). The additional introduction of year effects leaves our estimate nearly unchanged. This seems reasonable, as the tax share is—by construction—expressed in relative terms.

According to our preferred specification (Table 6 column 4) a hypothetical mining company (and sole tax payer) is associated with an 18 percentage points higher probability of an assassination. This translates to an average effect of about 1.1 percentage points, as the average tax share in

⁴⁵Note that we account for changes in ownership shares over time as well as for acquisitions.

the sample is 5.9 percent, which constitutes a 26 percent increase in the average probability of observing an event.⁴⁶

The estimated effects should be interpreted with care, as we cannot entirely rule out potential confounders. For example, journalists have an incentive to report about assassinations in association with large and renowned companies that are of most interest to readers. Since tax revenues are expected to be proportional to the value of projects owned by companies in the event country, reporting could simultaneously increase with the tax share. Reassuringly, the insignificant and positive effect of assassination events on the change in the tax shares (Δ Tax Share) across specifications reported in Table D.8 in the Appendix alleviates concerns about reverse causality.

6 Conclusion

Multinational companies are sometimes connected with human rights violations at the global periphery. In the absence of legal frameworks that govern corporate misbehaviour abroad, global civil society (e.g. activists, human rights groups, the media) may be some of the few institutions holding large multinationals accountable. In this paper, we evaluate the effect of publicising human right violations on the stock market value of the multinational companies connected with the abuse. In particular, we compile a unique database on 354 assassinations and extrajudicial killings of activists and link them to the publicly listed mining companies implicated in the events. We then combine this data with daily stock market returns of those companies and use Event Study Methodology to estimate the effect of the killings on the abnormal daily returns of the companies' stocks.

Our results show that killings of activists lead to statistically significant lower returns with a cumulative median loss of over USD 100 million in the 10 days following the event. We highlight the critical role of the media in making information available to a broader public in this context and show that the stock market's negative reaction to publicising the assassinations is more pronounced during days which are not dominated by other news-worthy events.

We believe that our results contribute to emerging work on the political economy of multinationals, and the political economy of human rights more broadly. Our findings show that informational campaigns by civil society have in fact an impact on multinational corporations. Being linked to human rights abuses can significantly influence a companies' stock market value.

⁴⁶The average probability to observe an assassination event in the sample is 4.16 percent.

7 Figures

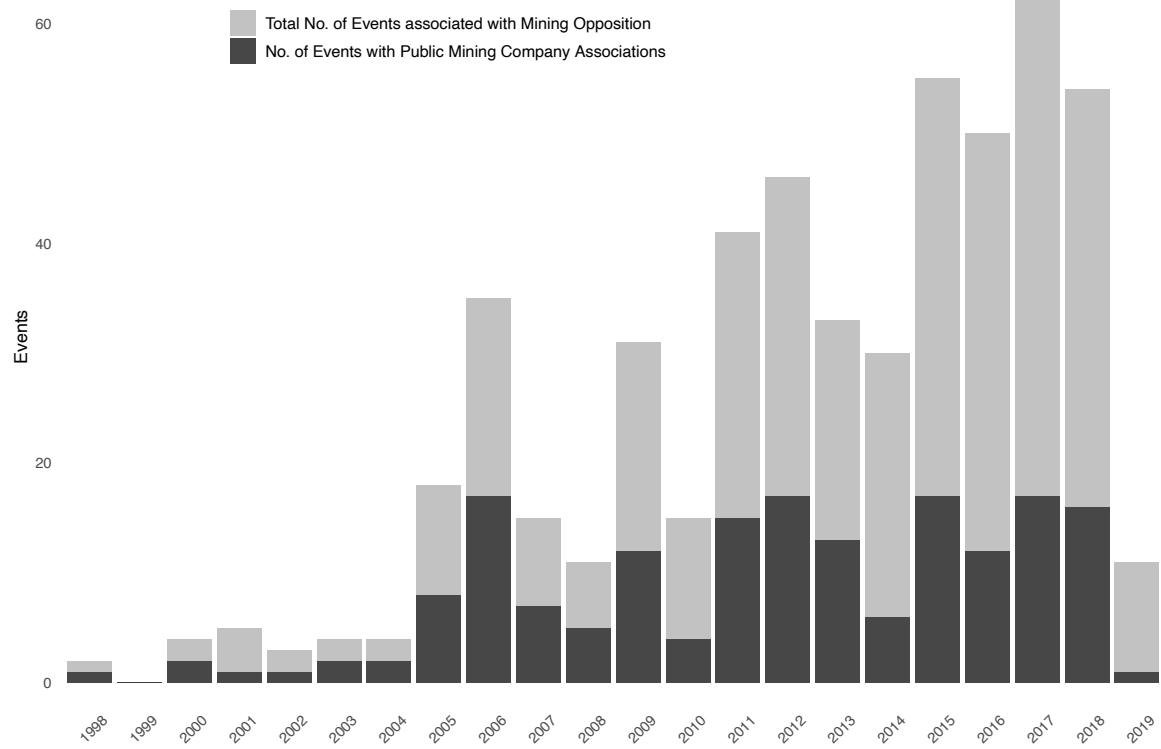


Figure 2: Distribution of Assassination Events over Time.

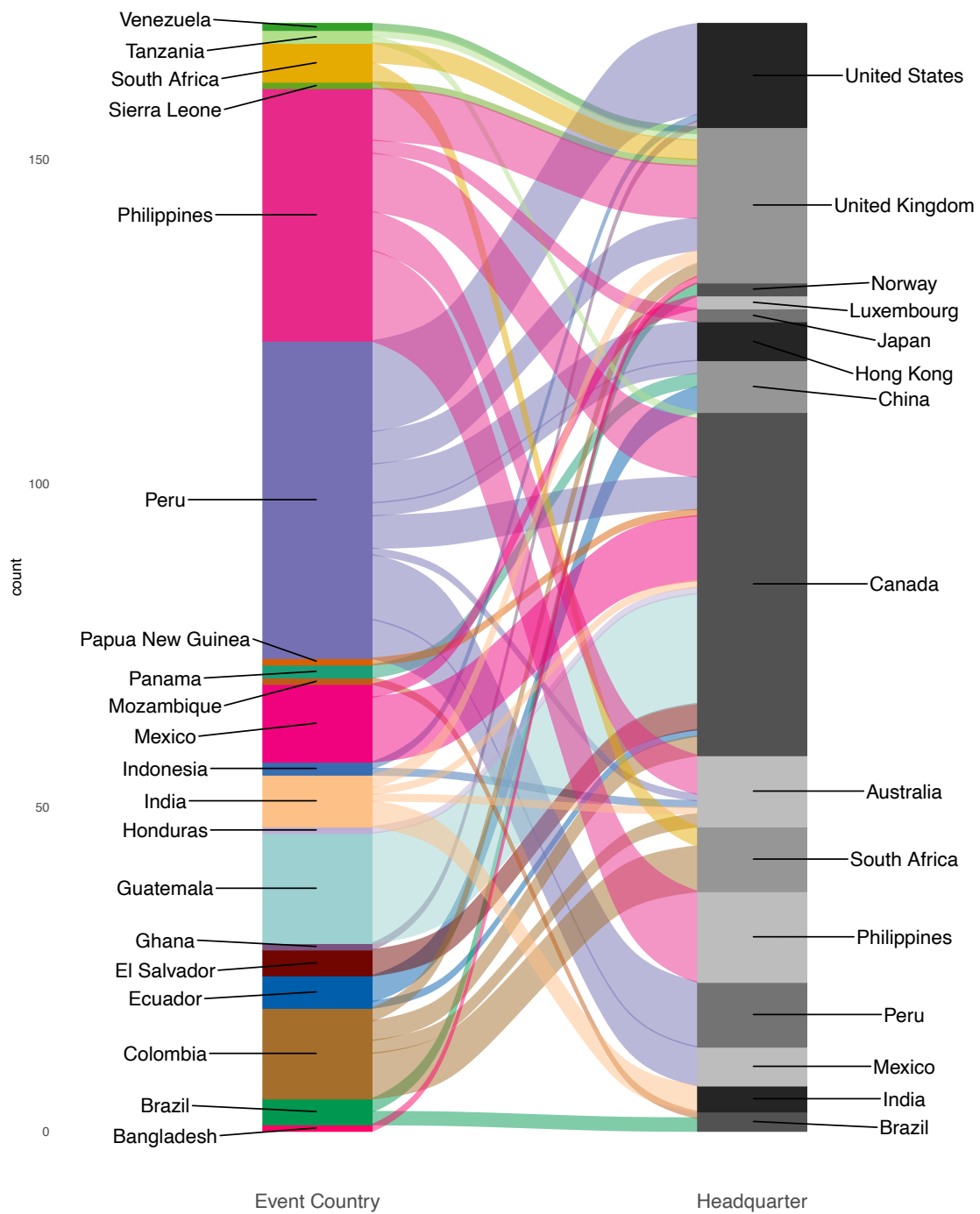


Figure 3: Event Country Activity and Company Headquarter Locations

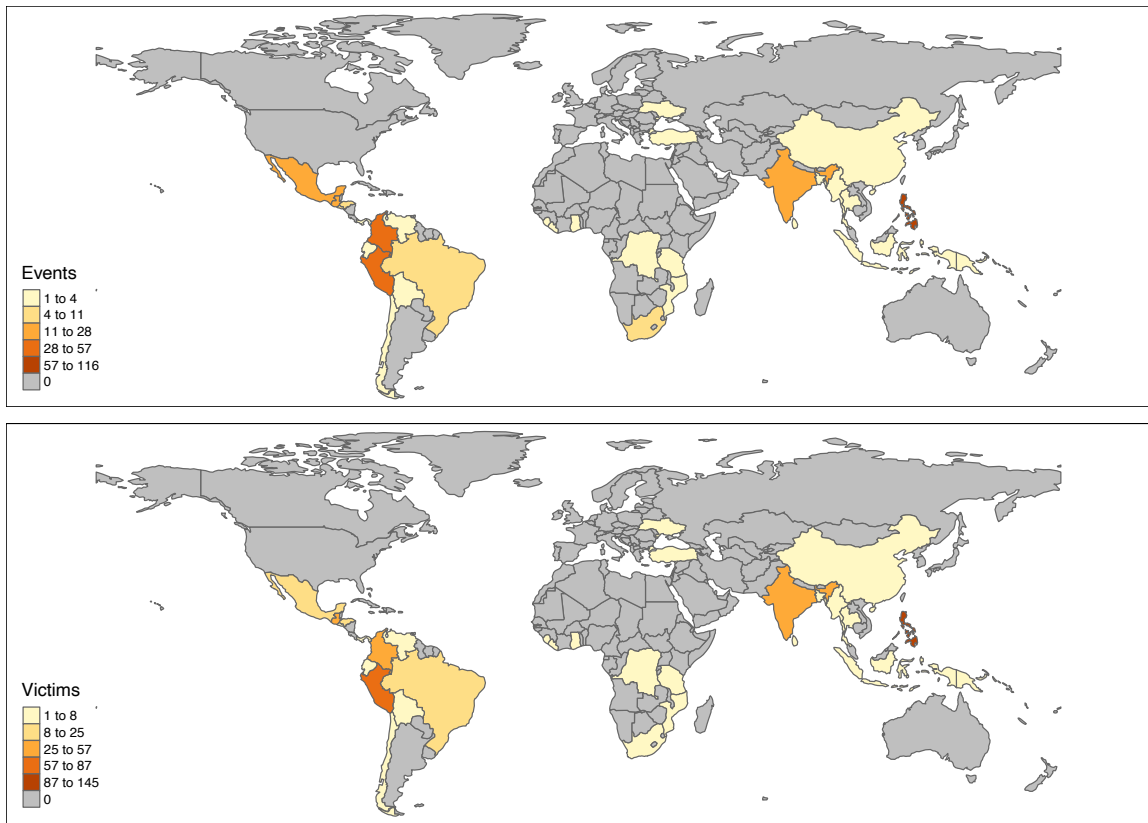


Figure 4: The Spatial Distribution of Assassination Events.

Ecuador indigenous leader found dead days before planned Lima protest

Shuar leader José Isidro Tendetza Antún missing since 28 November
Activists believe death linked to opposition to state Chinese mine project

Jonathan Watts, Latin America correspondent, and Dan Collyns in Lima

Sun 7 Dec 2014 09:59 AEDT

The body of an indigenous leader who was opposed to a major mining project in Ecuador has been found bound and buried, days before he planned to take his campaign to climate talks in Lima.

The killing highlights the violence and harassment facing environmental activists in Ecuador, following the confiscation last week of a bus carrying climate campaigners who planned to denounce president Rafael Correa at the United Nations conference.

The victim, José Isidro Tendetza Antún, a former vice-president of the Shuar Federation of Zamora, had been missing since 28 November, when he was last seen on his way to a meeting of protesters against the Mirador copper and gold mine. After a tip-off on Tuesday, his son Jorge unearthed the body from a grave marked "no name". The arms and legs were trussed by a blue rope.

- Event date
- Mining Project/Company
- "Assassination"/Violent death
- Mining opposition
- Name(s) and associations of the victim(s)

Figure 5: Extracting Events and Company Associations from NGO and Media Reports - Example Case

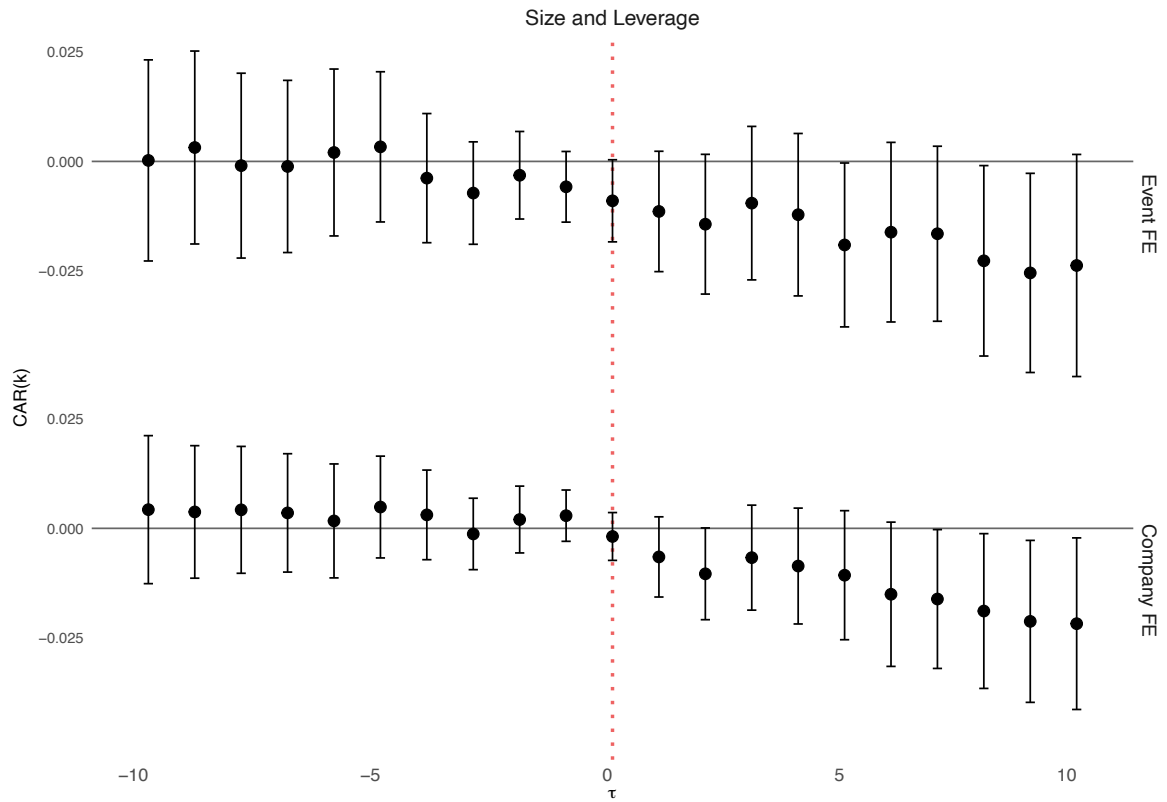


Figure 6: The Treatment Effect of Assassination Events on Mining Companies.

Notes: The coefficients when regressing the respective cumulative abnormal return (CAR) on an indicator for being tied to an assassination event is represented by the thick line (and dots). The horizontal axis label denotes the trading days before and after the event on $\tau = 0$. CARs are aggregated backwards before the event date and forwards starting with the event date. E.g. -5 refers to the CAR between -1 and -5 while 5 refers to the CAR between 0 and 5 . The top panel displays the point estimates for $\hat{\delta}$ when event fixed effects are included, the bottom panel estimates for the specification with company fixed effects. In total the coefficients of 42 regressions are displayed. 95% confidence intervals using robust standard errors clustered on the event-level are depicted.

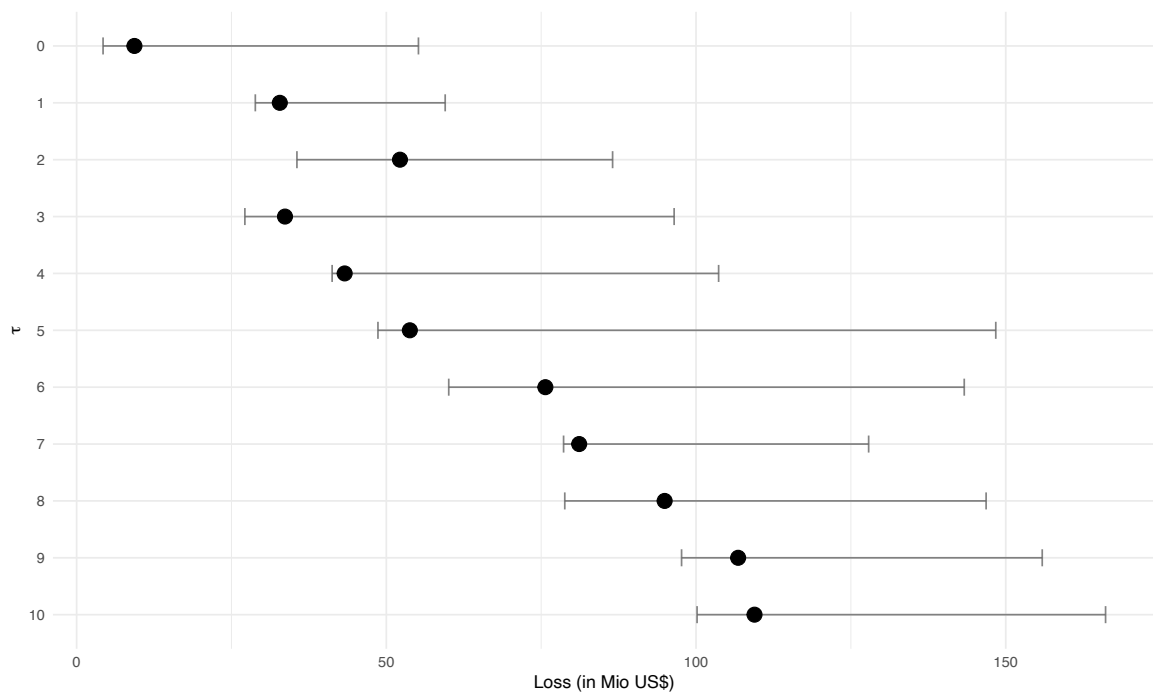


Figure 7: The estimated Economic Value of Assassination Events.

Notes: Dots correspond to the estimated loss in market capitalization of the median company for the baseline specification. The grey bars display the estimated minimum and maximum loss in market capitalization for the median company across specifications.

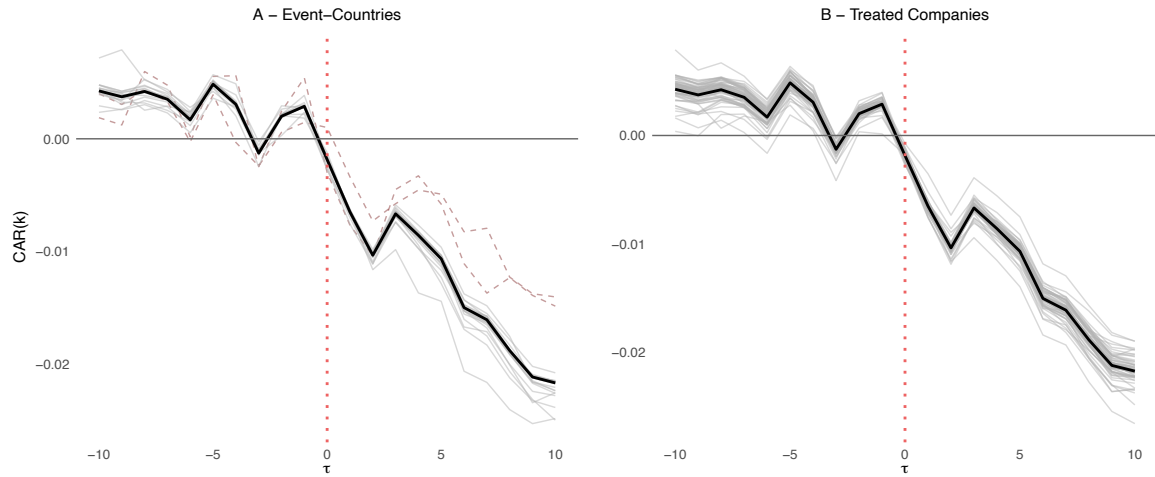


Figure 8: Robustness of Baseline Results - Leave-One-Out.

Notes: The thick black line in Panel-A and Panel-B corresponds to the baseline coefficient estimates for being tied to an assassination in the full sample. Panel-A presents the estimated coefficients when one country is consecutively dropped from the sample. The (red) dashed lines highlight the estimated coefficients when dropping events in the Philippines, respectively Peru from the sample. Panel-B displays the estimated coefficients when one treated company at a time is dropped from the sample.

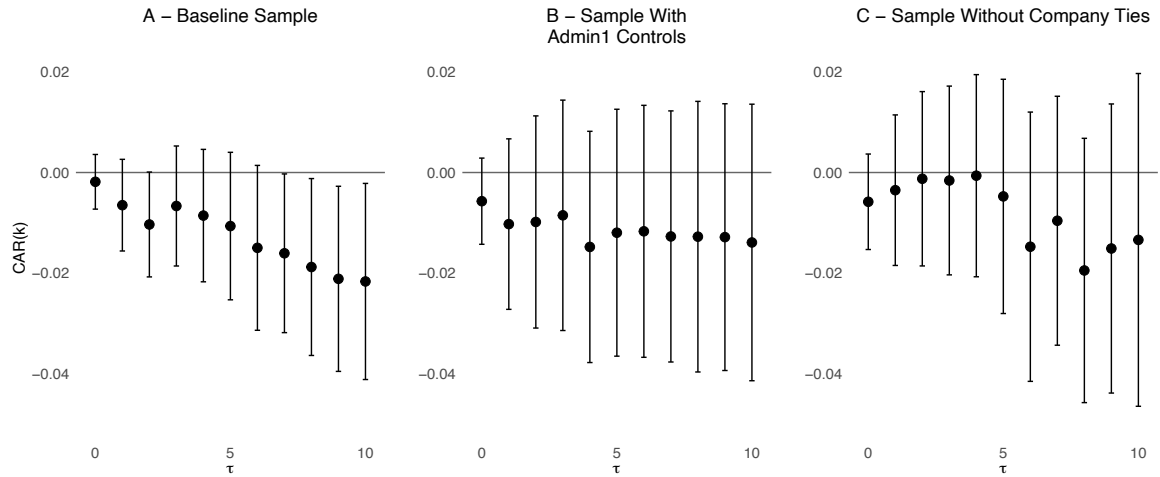


Figure 9: Vicinity vs. Media Ties.

Notes: The thick black lines correspond to the baseline coefficient estimates for being tied to an assassination. Panel-A presents the baseline sample estimates, while Panel-B presents the results when altering the control group to companies active in the Admin1 region of the assassination event. Panel-C shows the coefficient estimates for the sample with no public company associations; all companies within the same Admin1 region of the event are considered as treated, while all remaining companies in the event-country constitute the control group. 95% confidence intervals using robust standard errors clustered on the event-level are depicted.

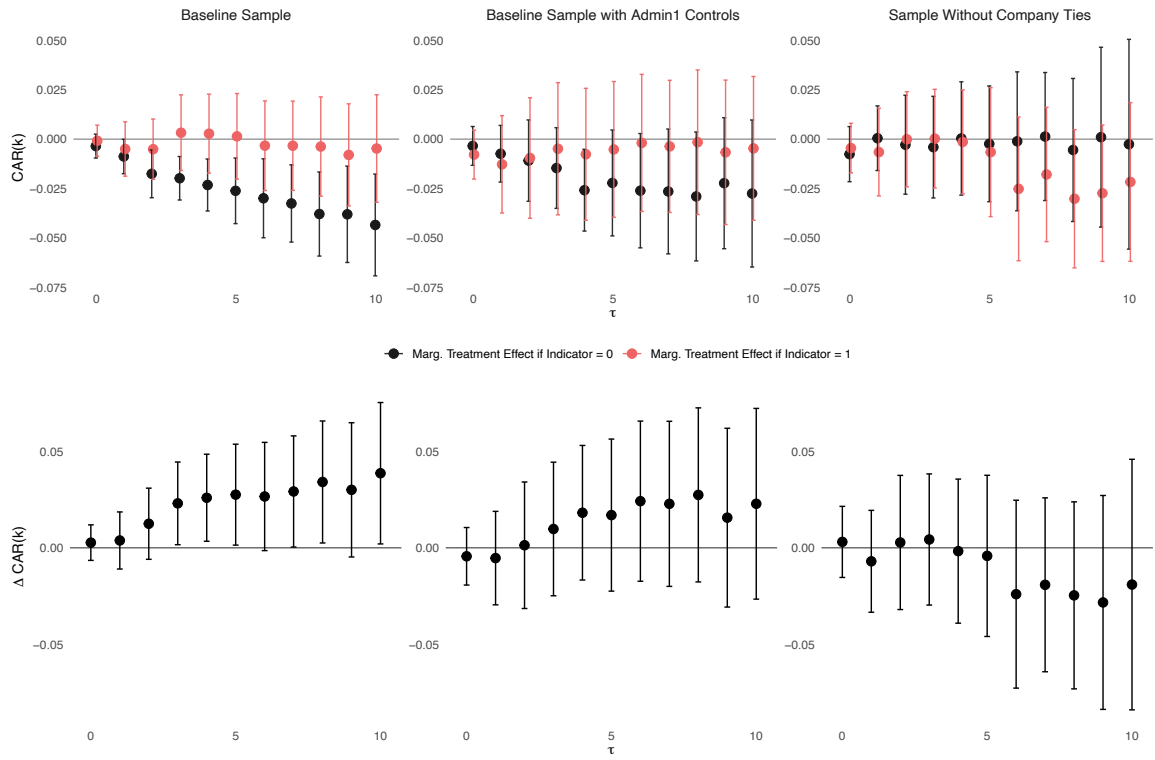


Figure 10: The Influence of News Pressure on the Event Day

Notes: The top panel displays the heterogeneous marginal treatment effect of assassination events on the respective cumulative abnormal return (CAR). The difference in treatment effects is presented in the bottom panel. The horizontal axis label denotes the trading days relative to the event day $\tau = 0$. CARs are forwards starting with the event date. E.g. 5 refers to the CAR between days 0 and 5. Columns present regression specifications with the assassination indicator interacted the indicator variable displayed in the column header. 95% confidence intervals using robust standard errors clustered on the event-level correspond to the error bars in the top panel and the ribbon in the bottom panel

8 Tables

Table 1: Assassination Summary Data

Country	Events		Victims	Assassination Attempts	Company-Event Pairs		Distinct Company Entities	
	Total	w/o Ties	Total	Total	Total	Public	Total	Public
Bangladesh	1	0	3	0	1	1	1	1
Bolivia	1	0	1	0	1	1	1	1
Brazil	11	7	11	0	4	4	2	2
Chile	1	1	1	0	0	0	0	0
China	1	0	4	0	1	0	1	0
Colombia	40	21	46	1	28	18	17	8
DR Congo	2	2	6	0	0	0	0	0
Ecuador	4	0	4	1	6	5	4	3
El Salvador	6	0	7	0	6	6	1	1
Gambia	1	0	2	0	1	0	1	0
Ghana	1	0	1	0	1	1	1	1
Guatemala	28	3	48	6	28	19	10	6
Honduras	9	4	12	1	6	2	6	2
India	25	15	57	0	12	9	10	7
Indonesia	4	1	5	0	5	3	5	3
Liberia	1	1	1	0	0	0	0	0
Mexico	21	4	25	0	20	17	12	9
Mozambique	1	0	1	0	1	1	1	1
Myanmar	4	1	4	0	4	0	4	0
Panama	1	0	2	0	2	2	2	2
Papua New Guinea	1	0	4	0	1	1	1	1
Peru	57	5	87	4	79	65	29	19
Philippines	116	57	145	1	85	57	43	27
Sierra Leone	1	0	1	0	1	1	1	1
South Africa	7	0	8	3	7	7	4	4
Sri Lanka	1	1	1	0	0	0	0	0
Tanzania	1	0	1	1	2	2	2	2
Thailand	3	2	3	0	1	0	1	0
Turkey	1	0	2	0	1	0	1	0
Ukraine	1	1	1	0	0	0	0	0
Venezuela	2	1	2	0	2	2	2	2
World	354	127	496	18	306	224	147	87

Notes: Events "w/o Ties" refer to events for which reports established a connection between the assassination (attempt) and the victim's opposition to mining, but no specific mining project or company was mentioned. The "Distinct Company Entities" entries correspond to the number of unique companies associated with assassination events in the respective country or on a world-wide scale.

Table 2: Summary Table - Financial Data

Group	Variable	Mean	St.dev.	Min	Max	Observations
Treatment	Raw return	0.0007	0.0029	-0.0041	0.0233	171
Control	Raw return	0.0011	0.0038	-0.0593	0.0476	4692
Treatment	Abnormal return	-0.0002	0.0010	-0.0044	0.0043	171
Control	Abnormal return	0.0000	0.0014	-0.0317	0.0149	4692
Treatment	Size	15.2984	2.6828	7.7807	20.6501	166
Control	Size	12.9156	3.2477	4.3307	20.6965	4512
Treatment	Leverage	0.0022	0.0019	0.0000	0.0072	165
Control	Leverage	0.1907	0.5757	0.0000	16.8088	4124
Treatment	Profitability	-0.0002	0.0046	-0.0440	0.0063	162
Control	Profitability	-0.2659	2.2810	-51.3538	17.9823	4389

Notes: Raw and abnormal returns for each security are previously averaged over the period from $\tau = -280$ to $\tau = +20$. Firm characteristics - i.e. size, leverage, and profitability - are based on the values in the event year.

Table 3: The Effect of Assassinations on Stock Returns

	Mean	SD	p-value			
			Normality	BMP	ADJ-BMP	GRANK
CAR0to0	-0.0009	0.0034	0.789	0.406	0.432	0.700
CAR0to1	-0.0066	0.0047	0.161	0.067	0.083	0.217
CAR0to2	-0.0074	0.0058	0.202	0.102	0.122	0.140
CAR0to3	-0.0040	0.0067	0.547	0.186	0.211	0.048
CAR0to4	-0.0061	0.0075	0.415	0.119	0.141	0.037
CAR0to5	-0.0078	0.0082	0.344	0.137	0.160	0.061
CAR0to6	-0.0104	0.0087	0.233	0.064	0.080	0.033
CAR0to7	-0.0132	0.0094	0.160	0.032	0.043	0.016
CAR0to8	-0.0148	0.0099	0.135	0.027	0.037	0.011
CAR0to9	-0.0201	0.0105	0.055	0.013	0.019	0.001
CAR0to10	-0.0200	0.0110	0.070	0.023	0.031	0.004

Notes: The number of company-event pairs N is 167. The respective average cumulative abnormal return (CAR) and its standard deviation (SD) is presented in columns 1 and 2 (c. equations (6) and (7) in Section 3.1). A minimum of 8 trading days during the *event window* from 0 to 10 is required. The *estimation window* spans from day -280 to -30 with a minimum of 200 trading days. Columns 3 - 6 show the p -value of the respective test-statistic. For details on the applied test-statistics see Appendix A.1.

Table 4: The Effect of Assassination Events on Institutional Investor Holdings

Dep. Variable:	IO	Institutional Investor Type						
		Banks	Insurance Comp.	Investment Comp.	Investment Advisors	Pesion Funds & Endowment	Hedge Funds	Top 5
Assassination	0.0010 (0.0047)	0.0002 (0.0002)	-0.0001 (0.0000)	-0.0005 (0.0017)	0.0054 (0.0041)	-0.0006 (0.0005)	-0.0034*** (0.0010)	0.0053 (0.0037)
Size and Leverage	X	X	X	X	X	X	X	X
Company FE	X	X	X	X	X	X	X	X
Company-specific quadratic time trend	X	X	X	X	X	X	X	X
R-squared	0.674	0.306	0.578	0.599	0.671	0.561	0.333	0.490
Observations	4136	4136	4136	4136	4136	4136	4136	4136
Mean	0.2484	0.0008	0.0006	0.0539	0.1588	0.0137	0.0206	0.1059

Notes: Robust standard errors clustered on the company-level in parentheses: *p<0.1, ** p<0.05, *** p<0.01.

Table 5: EITI Tax Revenue Share Data

Country	Years	Observations	Mean	St.dev.	Min	Max	Assassinations
Colombia	5	45	0.1111	0.1096	0.0001	0.3378	5
Ghana	13	138	0.0942	0.1170	0.0000	0.4927	0
Guatemala	2	23	0.0870	0.2781	0.0000	0.9901	3
Honduras	3	15	0.2000	0.2023	0.0062	0.5156	0
Mozambique	7	213	0.0329	0.1138	0.0000	0.9311	0
Papua New Guinea	5	40	0.1250	0.1822	0.0000	0.6291	1
Peru	13	331	0.0393	0.0796	0.0000	0.7864	28
Philippines	5	144	0.0347	0.0661	0.0000	0.4379	7
Sierra Leone	11	132	0.0833	0.1026	0.0006	0.4671	1

Notes: The number of events corresponds to the assassination events that can be matched to both, private and publicly traded mining companies with EITI tax records.

Table 6: Tax Revenue Shares and the Likelihood to observe Assassinations

	Dependent Variable: Assassination			
	(1)	(2)	(3)	(4)
Tax share	0.138** (0.060)	0.174*** (0.067)	0.174** (0.069)	0.181** (0.071)
Country FE		X	X	
Year FE			X	
Country $\times$ Year FE				X
R-squared	0.006	0.051	0.080	0.143
Observations	1081	1081	1081	1081

Notes: Robust standard errors in parentheses: *p<0.1, ** p<0.05, *** p<0.01.

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A Technical Appendix

A.1 Test Statistics

For ease of notation (and without loss of generality), we present test statistics for one particular aggregation period from τ_1 to τ_2 in this section, where $T_1 < \tau_1 \leq \tau_2 \leq T_2$.⁴⁷

Normality

Following MacKinlay (1997), the null hypothesis H_0 of no event effect under the assumption of normally distributed security returns and absence of clustering can be tested using

$$\theta_1 = \frac{\overline{CAR}}{\sigma(\overline{CAR})} \sim N(0, 1), \quad (\text{A.1})$$

with $\overline{CAR}$ and $\sigma(\overline{CAR})$ defined in (6) and (7).

BMP

Given the estimated abnormal and cumulative abnormal returns and their sample variance in (3)-(3), the scaled abnormal (SAR) and cumulative abnormal (SCAR) returns during the *event window* $\tau = T_1 + 1, \dots, T_2$ are defined as:⁴⁸

$$SAR_{i,e,\tau} = \frac{\widehat{AR}_{i,e,\tau}}{\sigma(\widehat{AR}_{i,e,\tau})} \quad (\text{A.2})$$

$$SCAR_{i,e} = \frac{\widehat{CAR}_{i,e}}{\sigma(\widehat{CAR}_{i,e})}. \quad (\text{A.3})$$

Boehmer et al. (1991) define the following test-static:

$$t_{BMP} = \frac{\overline{SCAR}\sqrt{N}}{\sigma(SCAR_{i,e})}, \quad (\text{A.4})$$

where $\overline{SCAR}$ constitutes the average scaled abnormal return on event day τ and $\sigma(SCAR_{i,e})$ the cross-sectional standard deviation of the SCAR.⁴⁹

$$\overline{SCAR} = \frac{1}{N} \sum_{j=1}^N SCAR_{j,e} \quad (\text{A.5})$$

$$\sigma(SCAR_{i,e}) = \sqrt{\frac{1}{N-1} \sum_{j=1}^N (SCAR_{j,e} - \overline{SCAR})^2}. \quad (\text{A.6})$$

The rescaling of the SCARs by the cross-sectional standard deviation makes the BMP t -statistic robust to event-induced volatility.

ADJ-BMP

Kolari and Pynnönen (2010) relax the assumption of no clustering by allowing for covariance between the SARs. Under the assumption of equal variance of SARs, the authors show that the “true”

⁴⁷This allows us to drop the suffix (τ_1, τ_2) .

⁴⁸Note that the definition for SARs is equivalent during the *estimation window* $\tau = T_0 + 1, \dots, T_1$.

⁴⁹Note that (A.4)-(A.6) are equivalently calculated for the SAR.

cross-sectional variance of the SARs in this setting boils down to:

$$s^2(SAR_{i,e}) = \frac{\sigma^2(SAR_{i,e})}{N} (1 + (N-1)\bar{r}), \quad (\text{A.7})$$

where $\sigma^2(SAR_{i,e})$ is given in (A.6) and $\bar{r}$ is the average of the sample cross-correlations of the ARs during the *estimation window*. Using the variance formula in (A.7) the adjusted BMP (ADJ-BMP) t -statistic is:

$$t_{ADJ-BMP} = \frac{\overline{SAR}}{s(SAR_{i,e})} = \frac{\overline{SAR}\sqrt{N}}{\sigma(SAR_{i,e})\sqrt{1 + (N-1)\bar{r}}} = t_{BMP} \sqrt{\frac{1 - \bar{r}}{1 + (N-1)\bar{r}}} \quad (\text{A.8})$$

The test statistic is equivalent for SCARs under the assumption of the square-root rule of the standard deviation of returns over different return periods (s. Kolari and Pynnönen, 2010, p. 4003).

GRANK

Kolari and Pynnönen (2011) re-standardize the SCARs defined in (A.3) using the cross-section standard deviation of the SCARs defined in (A.6) to transform the SCAR to a random variable with zero mean and unit variance just as the other SARs defined in (A.2):⁵⁰

$$SCAR_{i,e}^* = \frac{SCAR_{i,e}}{\sigma(SCAR_{i,e})}. \quad (\text{A.9})$$

This allows Kolari and Pynnönen (2011) to define the generalized standardized abnormal return ($GSAR_{i,e,\tau}$) as:

$$GSAR_{i,e,\tau} = \begin{cases} SCAR_{i,e}^* & \text{for } \tau = \tau_1, \dots, \tau_2 \\ SAR_{i,e,\tau} & \text{for } \tau = T_0 + 1, \dots, T_1. \end{cases} \quad (\text{A.10})$$

Intuitively, the CAR period is treated as if there was only one day, the “cumulative return day” at $\tau = 0$ (Kolari and Pynnönen, 2011). The demeaned standardized abnormal ranks ($U_{i,e,\tau}$) of the GSARs are:

$$U_{i,e,\tau} = \frac{\text{Rank}(GSAR_{i,e,\tau})}{T+1} - \frac{1}{2}, \quad (\text{A.11})$$

where $\tau \in \mathcal{T} = \{T_0 + 1, \dots, T_1, 0\}$ and T is equal to the length of the estimation window plus the “cumulative return day”, i.e. $T = L_1 + 1 = T_1 - T_0 + 1$.

Since $U_{i,e,\tau}$ constitutes the demeaned rank of the GSAR, the null hypothesis of having no mean event effect, i.e. $H_0 : E[\overline{CAR}] = 0$, is equal to the expected rank of the GSAR being equal to zero for all company-event pairs on the “cumulative return day” ($E[U_{i,e,0}] = 0$). Kolari and Pynnönen (2011) show that the t -statistic for testing this null hypothesis is:

$$t_{GRANK} = Z \left(\frac{T-2}{T-1-Z^2} \right)^{\frac{1}{2}}, \quad (\text{A.12})$$

where

$$Z = \frac{\overline{U}_0}{\sigma(\overline{U})} \quad (\text{A.13})$$

⁵⁰In case of event-day clustering, it may be preferable to use the cross-correlation robust standard deviation $s^2(SCAR_{i,e})$. Kolari and Pynnönen (2010) note however that this substitution should not substantially alter the results for rank tests (s. footnote 7 on p. 4008).

with

$$\sigma(\overline{U}) = \sqrt{\frac{1}{T} \sum_{t \in \mathcal{T}} \frac{N_t}{N} \overline{U}_t^2} \quad (\text{A.14})$$

$$\overline{U}_\tau = \frac{1}{N_\tau} \sum_{j=1}^N U_{i,e,\tau}, \quad (\text{A.15})$$

where N_τ is the number of non-missing (valid) GSARs available at $\tau \in \mathcal{T} = \{T_0 + 1, \dots, T_1, 0\}$ and N is the number of all company-event pairs.

B Data Appendix

B.1 Assassination Dataset

In this appendix we describe in detail the compilation and coding of assassination events. The list of 354 extra-judicial killings of mining activists was retrieved from a range of sources that can broadly be categorized into:

1. We obtain information from NGOs and human rights associations such as “Global Witness” and “Amnesty International”, “Front Line Defenders” or “Bulatlat”.
2. We use international full-text newspaper archives (e.g. Gale full-texts collections of the International Herald Tribune and Associated Press wire archives) and prominent APIs (e.g. Guardian) to locate events via algorithmic searches. Specifically, we query the APIs and news archives for articles that contain a combination of “activist” keywords (activist, campaigner, indigenous, etc. and additionally variations of mining) and “assassination” keywords (kill, assassin, abduct, etc.). Both keyword lists were chosen semi-automatically by looking up cosine similarities from the *web2vec* word vectors pre-trained on the Google News data set (c. for instance Keith et al., 2017).⁵¹ The (deduplicated) list of returned articles is then manually inspected for relevant events. Note that we also experimented with training text classification models to automatically detect relevant articles. The specificity of our events in combination with the infrequent reporting, however, does not allow for the construction of a sufficient training corpus. Moreover, the data collection process revealed that many assassination events are covered by local newspapers or NGO reports, usually not available in news archives and APIs. These supplementary sources are described below.
3. We search local newspapers such as “La Republica” in Peru, “El Universo” in Ecuador, “El Pais” in Mexico or “El Espectador” in Colombia.
4. We rely on published books (e.g. Holden and Jacobson, 2012; Doyle and Whitmore, 2014) and studies (e.g. Imai et al., 2017; Spohr, 2016; Hamm et al., 2013). These sources often provide supplementary information on cases such as event classifications - i.e. mining, deforestation - and mining project/company associations. For instance, Holden and Jacobson (2012) provide a list of mining projects and their owners at the time in chapter 2 that can be matched with the mining projects mentioned in association with killings of anti-mining activists in chapter 5.

After locating assassination events of opposition leaders, indigenous and tribal leaders, and local environmentalists we ensure that the event is indeed linked to the victim’s opposition to a mining project, i.e. we require at least one source to state that opposition to mining is the (suspected) reason for the attack. In particular, for 211 of the 565 killings of activists we collected we are not able to establish a link to mining opposition. These 211 cases either comprise assassinations in relation to other sectors such as logging, pipelines, and hydro dams or the source articles provides no conclusive information on the characteristics of the victim’s activism. Next, we establish company “ties” for the 354 “mining related” events. We implement the following matching procedure:

⁵¹The Google News data set comprises about 100 billion words. The pre-trained *web2vec* word vectors can be found here: <https://code.google.com/archive/p/word2vec/>.

1. If a mining company is named in at least one article we check if the reported company is publicly traded. As a convention, we consider only the “downstream” publicly traded companies for the case that the named mining company is not the global ultimate owner, except if the global corporate owner is specifically tied to the event in one of the articles. For instance, if the article states that the assassination is linked to a mining project owned by *AngloGold Ashanti*, a publicly traded mining company ultimately owned by *Anglo American*, we do not classify *Anglo American* as being “associated” with the event unless a source article specifically mentions *Anglo American* as well. Moreover, we cross-validate—to the best of our abilities—if the company was active in the country at the time of the event by inspecting—among others—annual reports. For the case that the named company is privately owned, we record the company name and do not further discern the ownership structure by private individuals.
2. If the stated mining company is not publicly traded, we examine if the company constitutes a subsidiary or joint venture of publicly traded companies at the time of the event by consulting—among others—company websites, annual reports, SEC documents and business registers. In case no company but a specific mining project could be identified, we rely on the aforementioned sources to establish the ownership structure of the mining project at the time of the event. In both cases, all owners are matched to the respective event. If a private company is the (partial) owner of a subsidiary/joint venture, the name of the company is recorded, not the name of the private owners of the company.

Apart from the company information, we hand-code (i) the precise event date, (ii) the name and number of the victims (iii) the geolocation of the assassination event⁵² (iv) the event “circumstances” (e.g. if the assassination attempt was successful or if it happened during a protest) (v) and—if known—the perpetrator (e.g. police, paramilitary forces, private security guards, hitmen).

⁵²For most assassination events, we are able to establish the exact assassination location. If the location is not known precisely, but only at the municipality level, we pick (approximately) the centroid of the municipality.

C Additional Figures

Broadcaster **gunned down** in the Philippines

October 19, 2011 1:35 PM EDT

*New York, October 19, 2011—A radio commentator and anti-mining tribal activist who was scheduled to launch a new radio station program in a few days was **gunned down** in the southern Philippines on October 14, news reports said.*

Roy Bagtikan Gallego was shot several times by men on a motorcycle as he was riding his motorcycle in Lianga town in Surigao del Sur province, in the southern Philippines, news reports said. The journalist was due to start a new block-time program with 92.7 Smile FM San Francisco, the reports said. Block-timing is a practice whereby a broadcaster leases air time from a radio station and is responsible for bringing in advertising money to cover the expenses of the program. A number of block-time commentators have been killed in the Philippines, according to CPJ research. In 2010, Gallego had hosted a similar program on DxFM San Francisco Radio, news reports said.

“Roy Bagtikan Gallego’s death must be investigated and the perpetrators prosecuted,” said Bob Dietz, CPJ’s Asia program coordinator. “Gallego’s death is emblematic of a much larger problem. In the Philippines, journalism and political activism are often conjoined, and the government must address the murders of journalists who use local media to take on controversial issues that threaten not only their lives but the strength of the nation’s media.”

Local police say they are investigating the death of Gallego, but have reached no conclusions on a possible motive and have not identified suspects, news reports said. Gallego, also a tribal leader of the Manobo tribe, had led the fight against small- and large-scale mining operators whose activities he claimed violated the rights of indigenous people in the region.

- Event date
- Mining opposition
- Mining Project/Company
- Name(s) and associations of the victim(s)
- “Assassination”/Violent death

Figure C.1: Example Case: Mining Opposition without Company Associations.

Notes: The source article can be found here: <https://cpj.org/2011/10/broadcaster-gunned-down-in-philippines/>

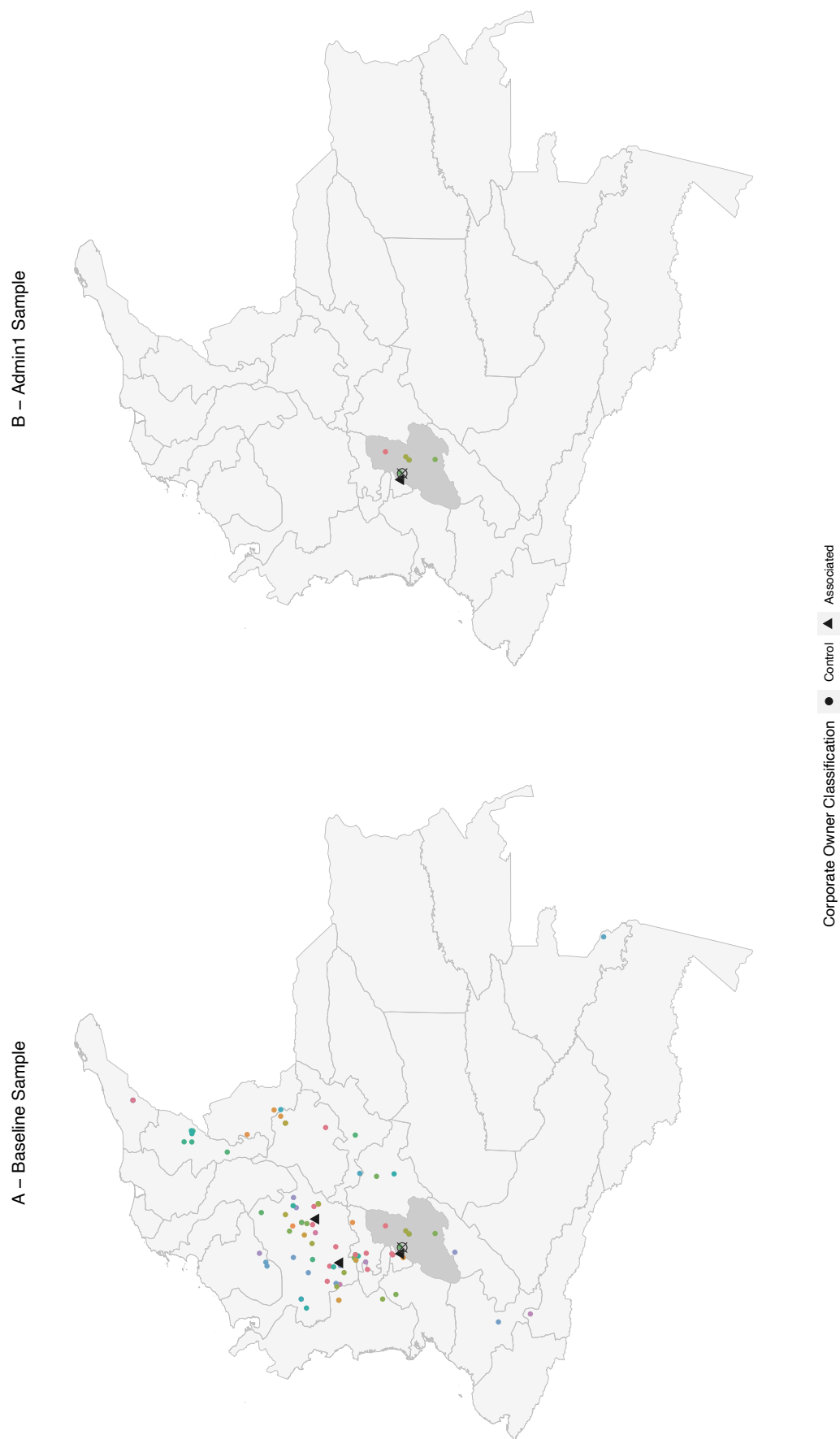


Figure C.2: The Construction of the Control Group - An Example Case from Colombia.

Notes: The map displays the Admin1 regions of mainland Colombia. The dark grey area corresponds to the Admin1 region, where the assassination event took place, while the location itself is marked by the black circle cross. Triangles (dots) correspond to mining projects owned by companies linked to the assassination event (or not), with colours differentiating corporate owner(s). Panel A displays all mining projects in the SNL database with ownership information in the event year (here: 2013). Panel B restricts the mining projects to the ones present in the Admin1 region of the assassination location.

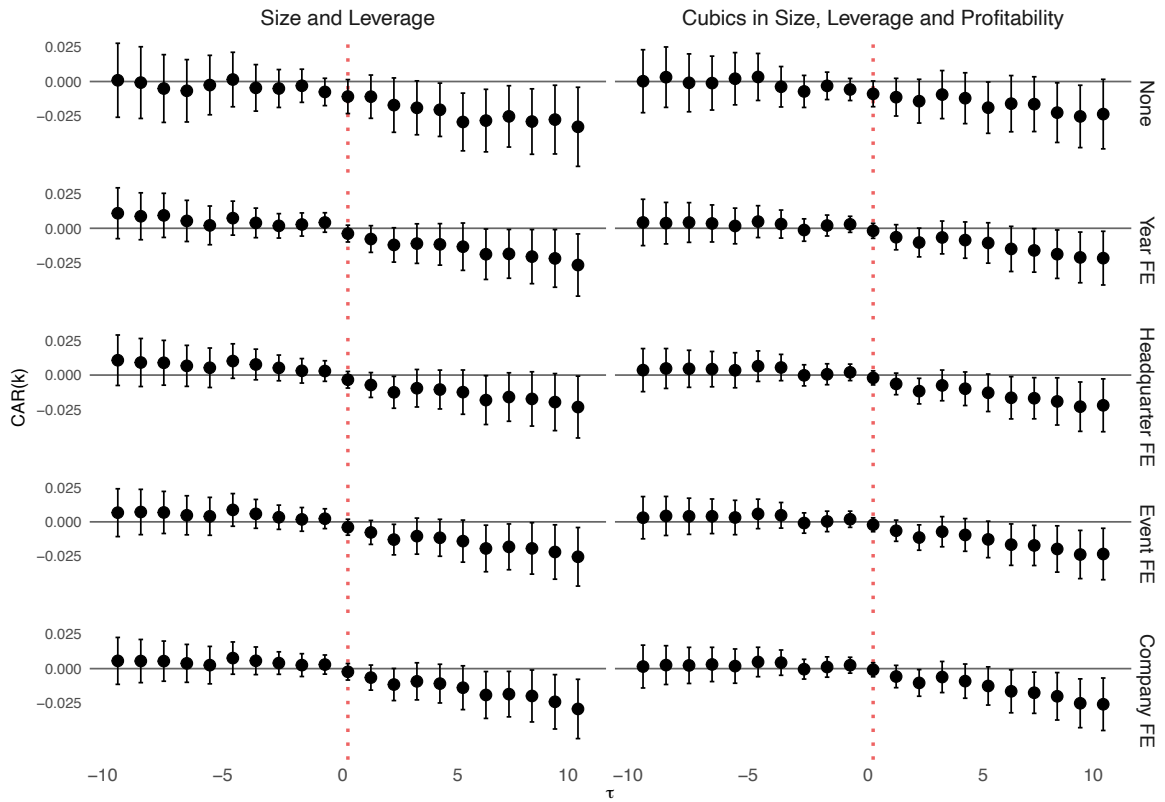


Figure C.3: The Treatment Effect of Assassination Events on Mining Companies.

Notes: The coefficients when regressing the respective cumulative abnormal return (CAR) on an indicator for being tied to an assassination event is represented by the thick line (and dots). The horizontal axis label denotes the trading days before and after the event on $\tau = 0$. CARs are aggregated backwards before the event date and forwards starting with the event date. E.g. -5 refers to the CAR between -1 and -5 while 5 refers to the CAR between 0 and 5 . Each cell corresponds to a different regression specification, with columns capturing control variable definitions and rows the inclusion of various fixed effects. In total the coefficients of 210 regressions are displayed. 95% confidence intervals using robust standard errors clustered on the event-level are depicted.

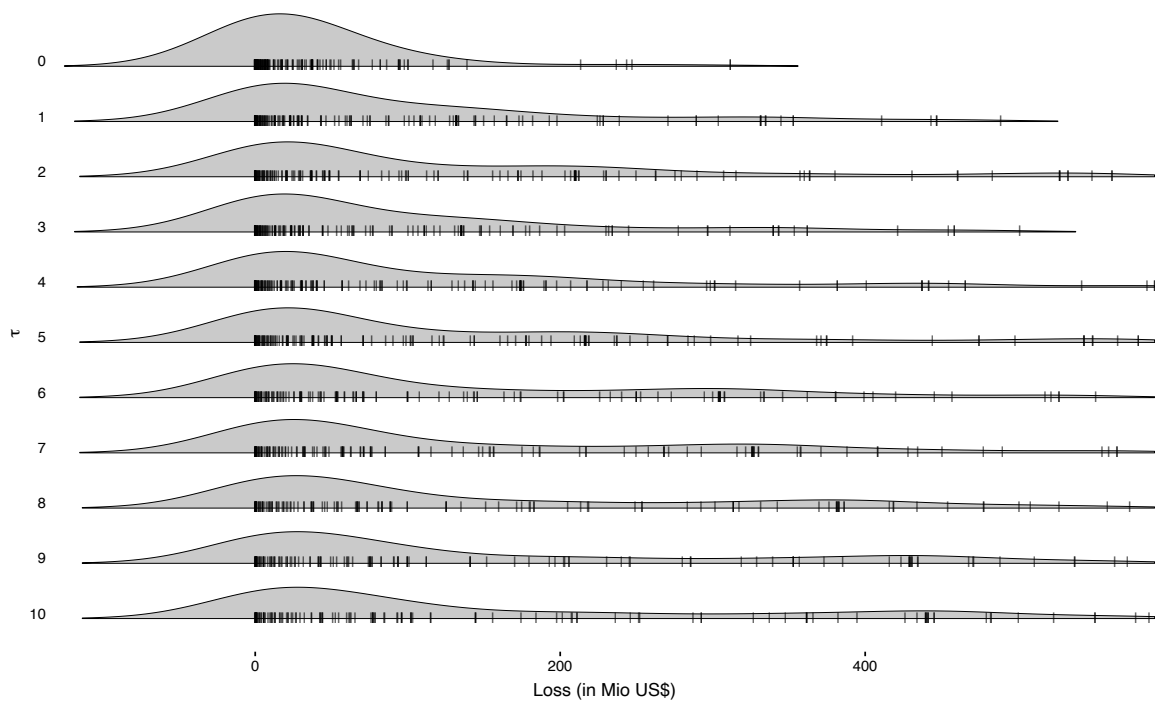


Figure C.4: The Distribution of the Economic Value of Assassination Events.

Notes: The distribution of market capitalization losses across companies for the baseline specification is displayed. For illustrative purposes losses above the 90th percentile are not displayed in Panel-B.

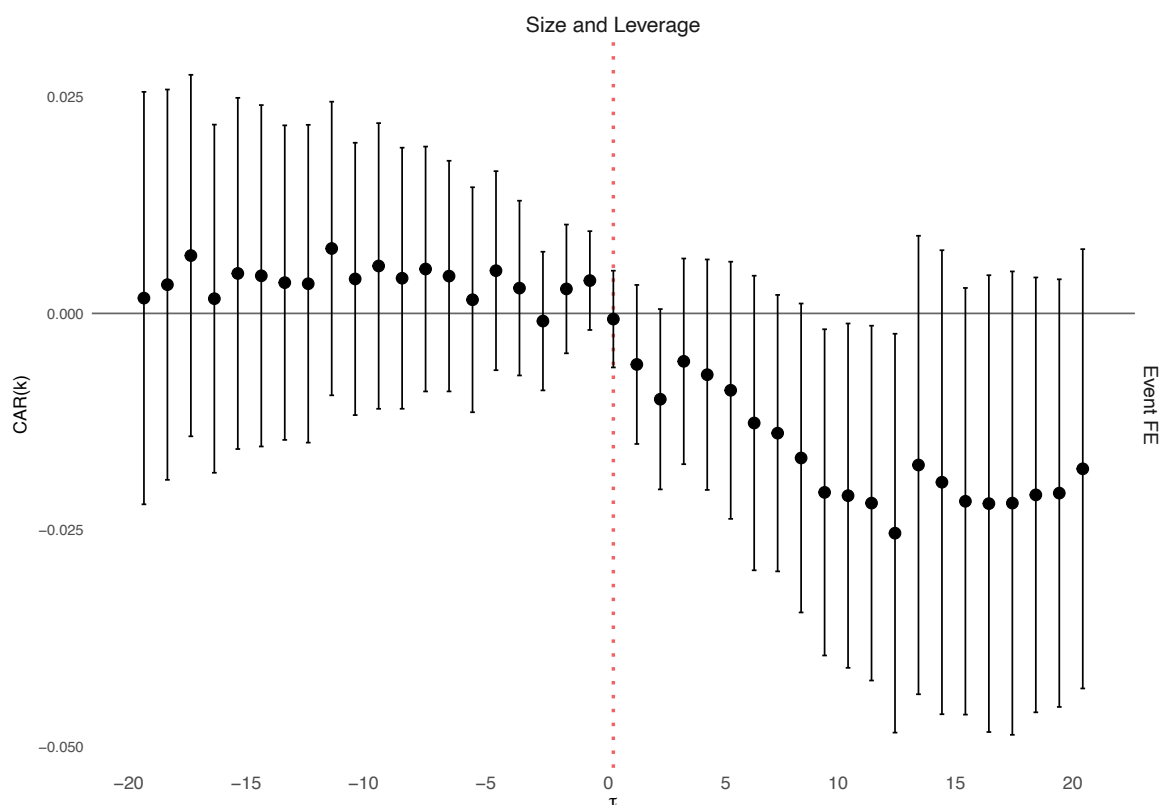


Figure C.5: The Treatment Effect of Assassination Events on Mining Companies - Wide Event Window.

Notes: The coefficients when regressing the respective cumulative abnormal return (CAR) on an indicator for being tied to an assassination event is represented by the thick line (and dots). Each dot corresponds to a separate regression coefficient estimate. The horizontal axis label denotes the trading days before and after the event on $\tau = 0$. CARs are aggregated backwards before the event date and forwards starting with the event date. E.g. -5 refers to the CAR between -1 and -5 while 5 refers to the CAR between 0 and 5 . 95% confidence intervals using robust standard errors clustered on the event-level are depicted.

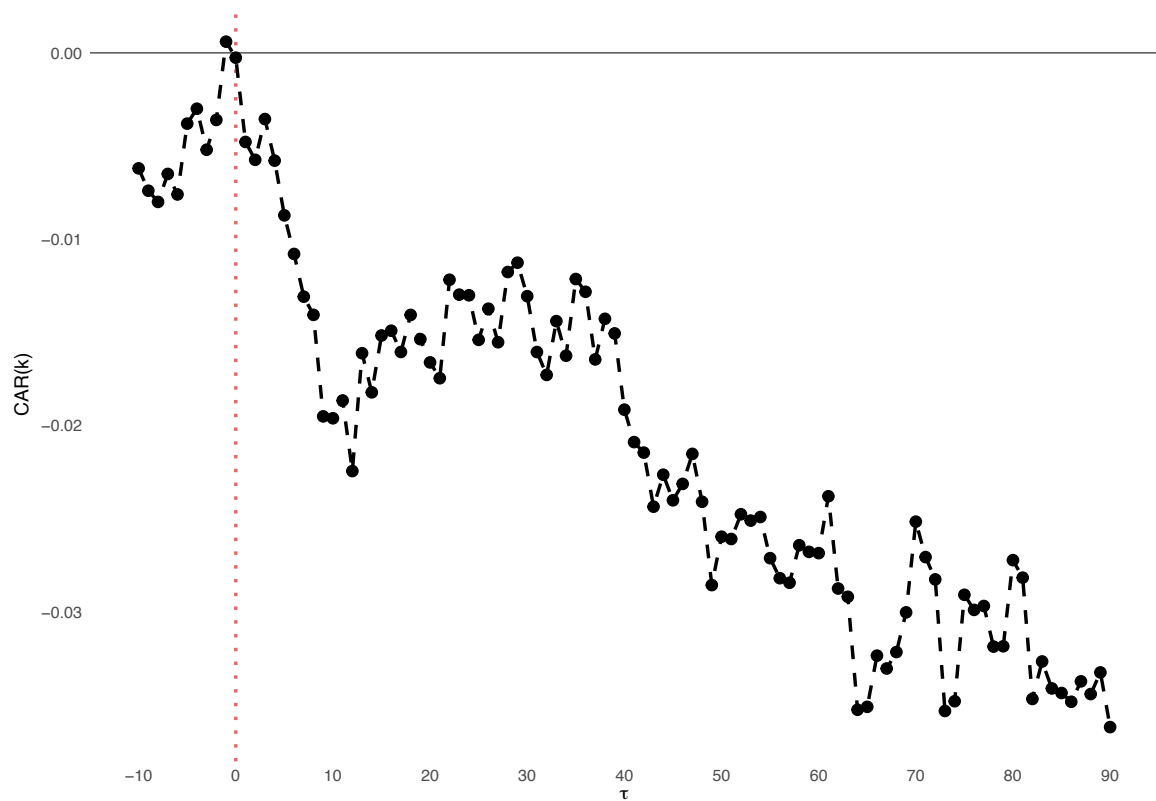


Figure C.6: The Long-Run Average Cumulative Abnormal Returns of Associated Companies.

Notes: Underlying the Market Model, the *average* cumulative abnormal return (CAR) of mining companies associated with assassination events are displayed. CARs are aggregated backwards before the event date and forwards starting with the event date. E.g. -5 refers to the CAR between -1 and -5 while 5 refers to the CAR between 0 and 5. Companies have to be traded 70 out of the 91 days following the event and 8 out 10 days prior to the event.

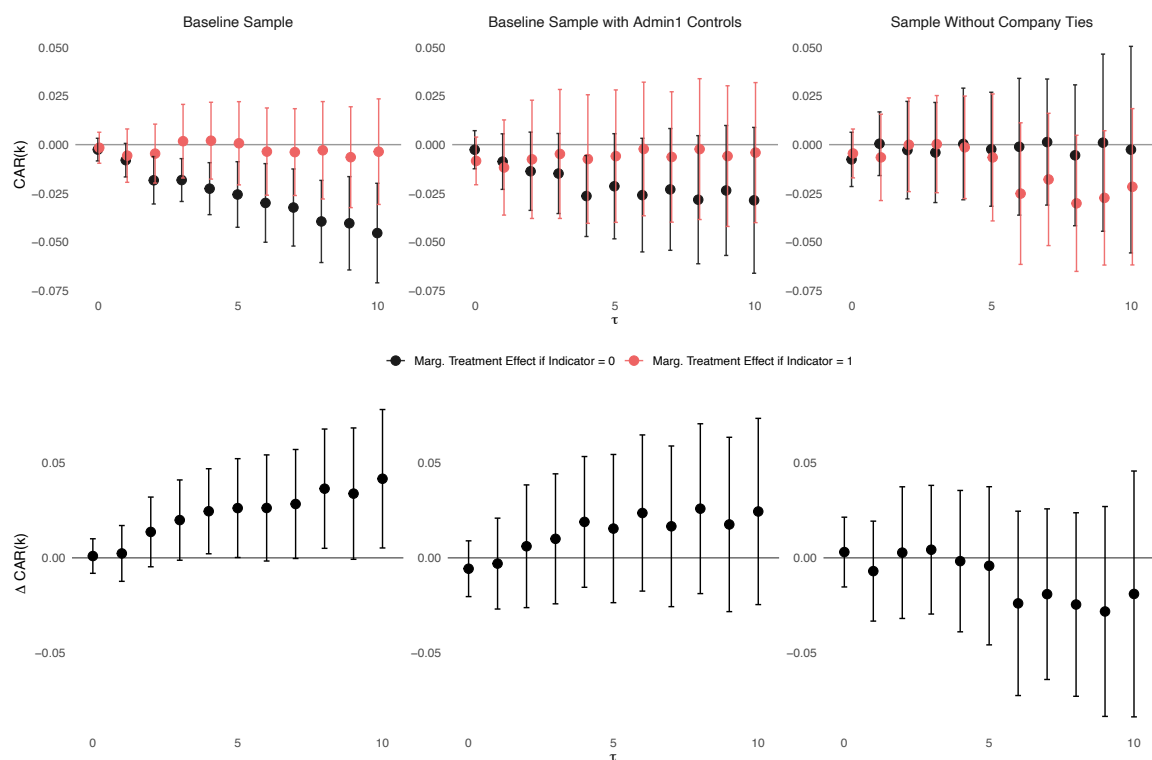


Figure C.7: The Influence of News Pressure on the Event Day - Robustness: Detrended

Notes: The top panel displays the heterogeneous marginal treatment effect of assassination events on the respective cumulative abnormal return (CAR). The difference in treatment effects is presented in the bottom panel. The horizontal axis label denotes the trading days relative to the event day $\tau = 0$. CARs are forwards starting with the event date. E.g. 5 refers to the CAR between days 0 and 5. Columns present regression specifications with the assassination indicator interacted the indicator variable displayed in the column header. 95% confidence intervals using robust standard errors clustered on the event-level correspond to the error bars in the top panel and the ribbon in the bottom panel

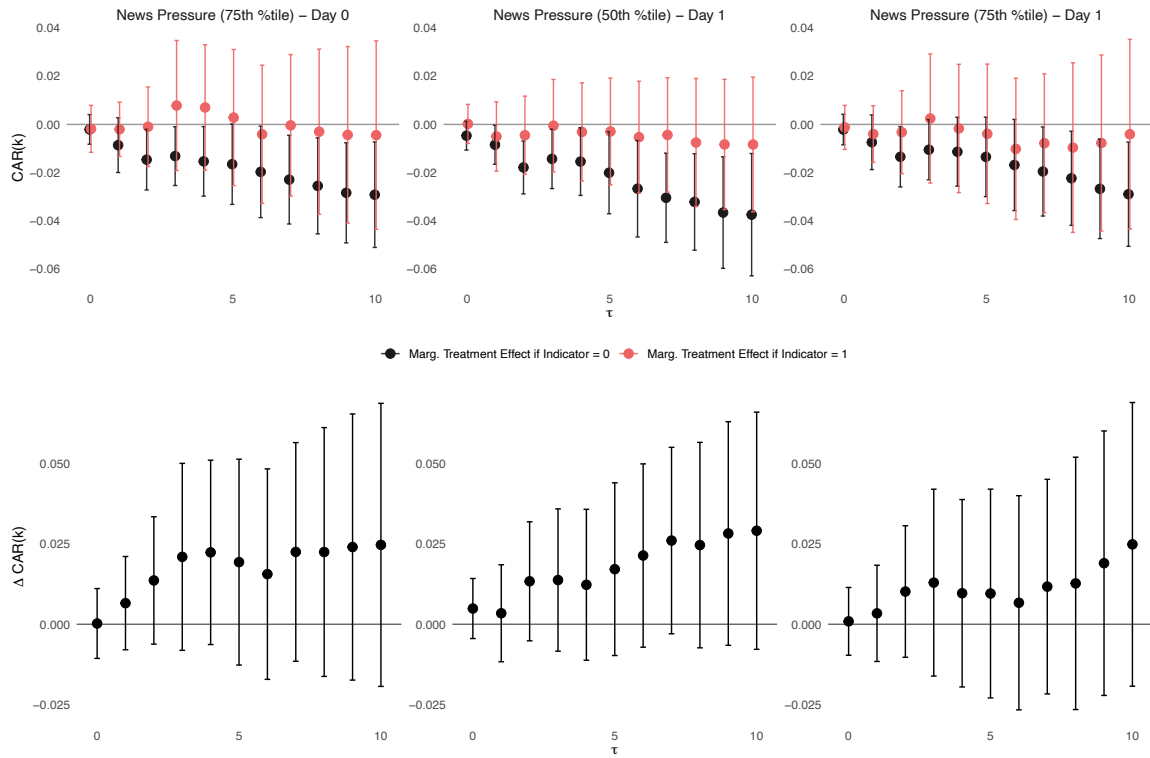


Figure C.8: The Influence of News Pressure - Robustness: 75th Percentile and Day 1

Notes: The top panel displays the heterogeneous marginal treatment effect of assassination events on the respective cumulative abnormal return (CAR). The difference in treatment effects is presented in the bottom panel. The horizontal axis label denotes the trading days relative to the event day $\tau = 0$. CARs are forwards starting with the event date. E.g. 5 refers to the CAR between days 0 and 5. Columns present regression specifications with the assassination indicator interacted the indicator variable displayed in the column header. 95% confidence intervals using robust standard errors clustered on the event-level correspond to the error bars in the top panel and the ribbon in the bottom panel

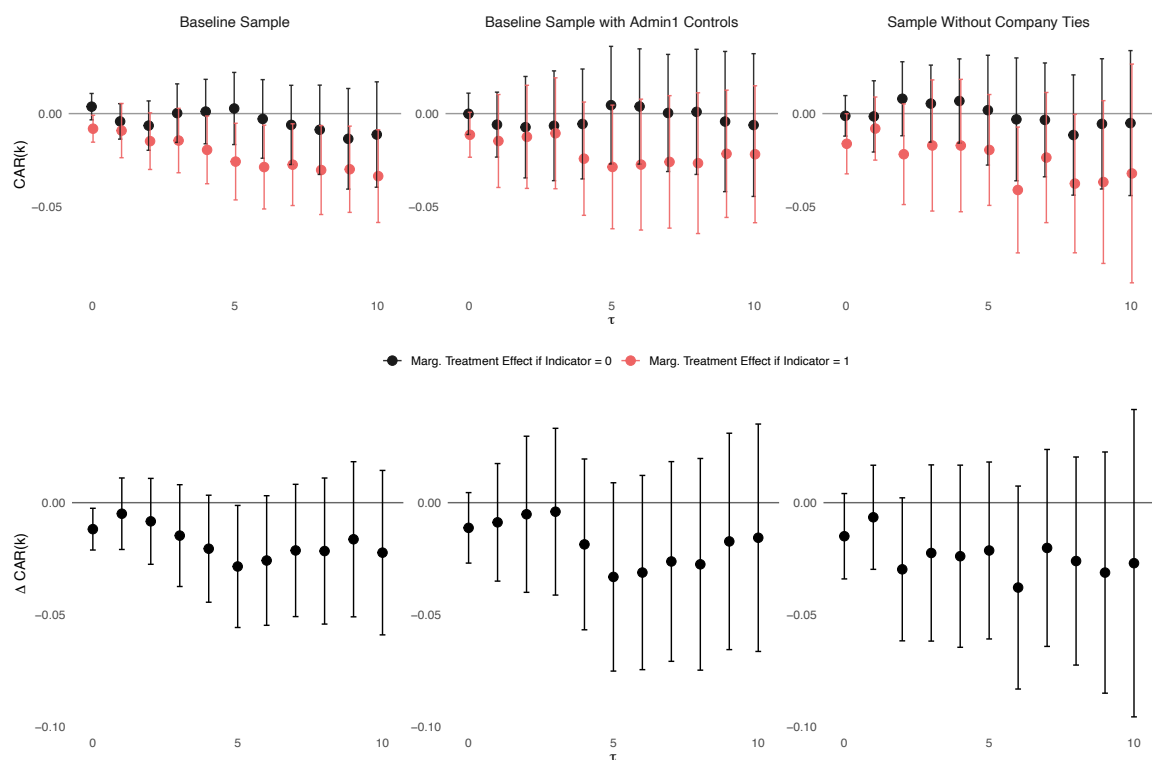


Figure C.9: The Influence of Oversight - EITI Membership

Notes: The top panel displays the heterogeneous marginal treatment effect of assassination events on the respective cumulative abnormal return (CAR). The difference in treatment effects is presented in the bottom panel. The horizontal axis label denotes the trading days relative to the event day $\tau = 0$. CARs are forwards starting with the event date. E.g. 5 refers to the CAR between days 0 and 5. Columns present regression specifications with the assassination indicator interacted the indicator variable displayed in the column header. 95% confidence intervals using robust standard errors clustered on the event-level correspond to the error bars in the top panel and the ribbon in the bottom panel

D Additional Tables

Table D.1: The Effect of Assassinations on Frequently Traded Companies

	Mean	SD	p-value			
			Normality	BMP	ADJ-BMP	GRANK
CAR0to0	-0.0002	0.0024	0.934	0.465	0.498	0.632
CAR0to1	-0.0054	0.0035	0.118	0.081	0.106	0.348
CAR0to2	-0.0047	0.0043	0.270	0.145	0.176	0.255
CAR0to3	-0.0035	0.0049	0.480	0.172	0.205	0.047
CAR0to4	-0.0038	0.0055	0.496	0.136	0.166	0.061
CAR0to5	-0.0040	0.0061	0.511	0.175	0.209	0.114
CAR0to6	-0.0062	0.0066	0.347	0.089	0.115	0.069
CAR0to7	-0.0092	0.0070	0.190	0.048	0.066	0.043
CAR0to8	-0.0110	0.0075	0.139	0.038	0.054	0.027
CAR0to9	-0.0153	0.0079	0.052	0.018	0.028	0.004
CAR0to10	-0.0155	0.0083	0.061	0.030	0.044	0.008

Notes: The number of company-event pairs N is 160. The respective average cumulative abnormal return (CAR) and its standard deviation (SD) is presented in columns 1 and 2 (c. equations (6) and (7) in Section 3.1). A minimum of 11 trading days during the *event window* from 0 to 10 is required. *The estimation window* spans from day -280 to -30 with a minimum of 225 trading days. Columns 3 - 6 show the p -value of the respective test-statistic. For details on the applied test-statistics see Appendix A.1.

Table D.2: Private Information and Pre-Trends

	Mean	SD	p-value			
			Normality	BMP	ADJ-BMP	GRANK
CAR-1to-1	0.0006	0.0034	0.870	0.782	0.783	0.706
CAR-1to-2	-0.0036	0.0048	0.447	0.236	0.237	0.231
CAR-1to-3	-0.0052	0.0058	0.370	0.157	0.158	0.129
CAR-1to-4	-0.0030	0.0068	0.656	0.412	0.413	0.240
CAR-1to-5	-0.0038	0.0076	0.618	0.520	0.521	0.303
CAR-1to-6	-0.0076	0.0083	0.360	0.487	0.488	0.234
CAR-1to-7	-0.0065	0.0090	0.470	0.562	0.563	0.292
CAR-1to-8	-0.0080	0.0095	0.401	0.413	0.415	0.157
CAR-1to-9	-0.0074	0.0101	0.462	0.569	0.570	0.216
CAR-1to-10	-0.0062	0.0106	0.560	0.801	0.802	0.364

Notes: The number of company-event pairs N is 170. The respective average cumulative abnormal return (CAR) and its standard deviation (SD) is presented in columns 1 and 2 (c. equations (6) and (7) in Section 3.1). A minimum of 8 trading days during the *event window* from -1 to -10 is required. *The estimation window* spans from day -280 to -30 with a minimum of 200 trading days. Columns 3 - 6 show the p -value of the respective test-statistic. For details on the applied test-statistics see Appendix A.1.

Table D.3: The effect of assassination events on stock prices - OLS regression results.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
CAR0to0	-0.0021 (0.0026)	-0.0020 (0.0026)	-0.0008 (0.0026)	-0.0019 (0.0028)	-0.0090* (0.0047)	-0.0039 (0.0029)	-0.0034 (0.0031)	-0.0023 (0.0030)	-0.0039 (0.0031)	-0.0109* (0.0062)
CAR0to1	-0.0065* (0.0039)	-0.0064 (0.0040)	-0.0057 (0.0041)	-0.0065 (0.0046)	-0.0114 (0.0069)	-0.0077* (0.0044)	-0.0072 (0.0046)	-0.0065 (0.0046)	-0.0078 (0.0049)	-0.0110 (0.0080)
CAR0to2	-0.0114** (0.0047)	-0.0117** (0.0047)	-0.0104** (0.0049)	-0.0104* (0.0053)	-0.0143* (0.0081)	-0.0129** (0.0056)	-0.0125** (0.0059)	-0.0116* (0.0059)	-0.0121* (0.0063)	-0.0172* (0.0100)
CAR0to3	-0.0072 (0.0056)	-0.0075 (0.0057)	-0.0061 (0.0057)	-0.0067 (0.0060)	-0.0095 (0.0088)	-0.0104 (0.0066)	-0.0096 (0.0069)	-0.0093 (0.0068)	-0.0111 (0.0073)	-0.0191* (0.0099)
CAR0to4	-0.0095 (0.0060)	-0.0099 (0.0062)	-0.0091 (0.0063)	-0.0086 (0.0067)	-0.0121 (0.0094)	-0.0116* (0.0068)	-0.0105 (0.0072)	-0.0109 (0.0071)	-0.0117 (0.0076)	-0.0206** (0.0098)
CAR0to5	-0.0128* (0.0067)	-0.0129* (0.0069)	-0.0126* (0.0070)	-0.0107 (0.0074)	-0.0190** (0.0094)	-0.0140* (0.0077)	-0.0124 (0.0081)	-0.0139* (0.0081)	-0.0134 (0.0087)	-0.0294*** (0.0106)
CAR0to6	-0.0166** (0.0076)	-0.0165** (0.0078)	-0.0165** (0.0079)	-0.0150* (0.0083)	-0.0161 (0.0103)	-0.0194** (0.0086)	-0.0182** (0.0090)	-0.0192** (0.0086)	-0.0189** (0.0093)	-0.0284** (0.0115)
CAR0to7	-0.0172** (0.0074)	-0.0168** (0.0076)	-0.0175** (0.0076)	-0.0161** (0.0080)	-0.0165 (0.0101)	-0.0182** (0.0084)	-0.0160* (0.0089)	-0.0185** (0.0084)	-0.0186** (0.0090)	-0.0254** (0.0113)
CAR0to8	-0.0197** (0.0085)	-0.0192** (0.0087)	-0.0201** (0.0087)	-0.0188** (0.0089)	-0.0226** (0.0110)	-0.0193** (0.0095)	-0.0173* (0.0100)	-0.0199** (0.0096)	-0.0206** (0.0100)	-0.0291** (0.0120)
CAR0to9	-0.0238*** (0.0089)	-0.0230** (0.0091)	-0.0252*** (0.0090)	-0.0212** (0.0093)	-0.0254** (0.0115)	-0.0219** (0.0100)	-0.0197* (0.0105)	-0.0241** (0.0100)	-0.0219** (0.0106)	-0.0277** (0.0126)
CAR0to10	-0.0234** (0.0095)	-0.0220** (0.0097)	-0.0259*** (0.0097)	-0.0217** (0.0099)	-0.0237* (0.0128)	-0.0254** (0.0108)	-0.0233** (0.0114)	-0.0293*** (0.0109)	-0.0267** (0.0114)	-0.0329** (0.0146)
Size and Leverage	X	X	X	X	X	X	X	X	X	X
Profitability						X	X	X	X	X
Cubic Terms						X	X	X	X	X
Headquarter FE		X					X			
Year FE			X							
Event FE				X					X	
Company FE					X					X
Observations	4177	4177	4175	4170	4090	4029	4029	4027	4022	3944
Clusters	154	154	152	147	153	154	154	152	147	153

Notes: Robust standard errors are clustered on the event-level. Standard errors in parentheses: *p<0.1, ** p<0.5, *** p<0.01.

Table D.4: The effect of assassination events on stock prices - OLS robustness checks.

	Excl. Attempts				Excl. Protests				Winsorized						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
CAR0to0	-0.0024 (0.0027)	-0.0024 (0.0028)	-0.0012 (0.0028)	-0.0018 (0.0029)	-0.0093* (0.0053)	-0.0034 (0.0032)	-0.0036 (0.0032)	-0.0014 (0.0031)	-0.0015 (0.0031)	-0.0094 (0.0067)	-0.0004 (0.0023)	-0.0005 (0.0024)	0.0007 (0.0023)	0.0005 (0.0025)	-0.0039 (0.0032)
CAR0to1	-0.0068 (0.0042)	-0.0068 (0.0043)	-0.0062 (0.0044)	-0.0064 (0.0050)	-0.0123 (0.0079)	-0.0091** (0.0036)	-0.0093** (0.0036)	-0.0074** (0.0036)	-0.0083** (0.0037)	-0.0096 (0.0073)	-0.0046* (0.0027)	-0.0046* (0.0027)	-0.0037 (0.0027)	-0.0045 (0.0029)	-0.0048 (0.0043)
CAR0to2	-0.0109** (0.0050)	-0.0111** (0.0051)	-0.0101* (0.0052)	-0.0095* (0.0057)	-0.0142 (0.0091)	-0.0126** (0.0053)	-0.0129** (0.0055)	-0.0105* (0.0053)	-0.0097* (0.0054)	-0.0156* (0.0093)	-0.0082** (0.0039)	-0.0086** (0.0039)	-0.0072* (0.0040)	-0.0076* (0.0041)	-0.0054 (0.0049)
CAR0to3	-0.0063 (0.0060)	-0.0067 (0.0061)	-0.0054 (0.0062)	-0.0054 (0.0065)	-0.0070 (0.0098)	-0.0069 (0.0071)	-0.0070 (0.0073)	-0.0039 (0.0070)	-0.0048 (0.0070)	-0.0081 (0.0101)	-0.0087** (0.0040)	-0.0092** (0.0040)	-0.0078* (0.0041)	-0.0083** (0.0041)	-0.0054 (0.0057)
CAR0to4	-0.0096 (0.0065)	-0.0100 (0.0067)	-0.0088 (0.0067)	-0.0076 (0.0072)	-0.0106 (0.0103)	-0.0062 (0.0080)	-0.0061 (0.0082)	-0.0030 (0.0081)	-0.0028 (0.0082)	-0.0061 (0.0105)	-0.0102** (0.0046)	-0.0109** (0.0047)	-0.0097** (0.0047)	-0.0095* (0.0049)	-0.0102* (0.0058)
CAR0to5	-0.0131* (0.0072)	-0.0134* (0.0074)	-0.0125* (0.0075)	-0.0102 (0.0080)	-0.0177* (0.0104)	-0.0125 (0.0087)	-0.0123 (0.0090)	-0.0092 (0.0090)	-0.0076 (0.0091)	-0.0173 (0.0132)	-0.0141*** (0.0052)	-0.0146*** (0.0052)	-0.0140*** (0.0053)	-0.0120** (0.0055)	-0.0132* (0.0062)
CAR0to6	-0.0184** (0.0080)	-0.0186** (0.0082)	-0.0177** (0.0083)	-0.0161* (0.0088)	-0.0162 (0.0110)	-0.0160 (0.0098)	-0.0160 (0.0099)	-0.0124 (0.0099)	-0.0122 (0.0101)	-0.0132 (0.0116)	-0.0172*** (0.0061)	-0.0175*** (0.0062)	-0.0175*** (0.0062)	-0.0165** (0.0064)	-0.0118 (0.0071)
CAR0to7	-0.0195** (0.0077)	-0.0196** (0.0079)	-0.0189** (0.0079)	-0.0172** (0.0085)	-0.0172 (0.0107)	-0.0181* (0.0096)	-0.0179* (0.0097)	-0.0149 (0.0095)	-0.0145 (0.0099)	-0.0160 (0.0117)	-0.0182*** (0.0061)	-0.0183*** (0.0062)	-0.0186*** (0.0061)	-0.0176*** (0.0064)	-0.0132* (0.0073)
CAR0to8	-0.0221** (0.0090)	-0.0220** (0.0092)	-0.0213** (0.0092)	-0.0203** (0.0095)	-0.0236** (0.0117)	-0.0161 (0.0114)	-0.0164 (0.0115)	-0.0122 (0.0112)	-0.0116 (0.0114)	-0.0213 (0.0133)	-0.0202*** (0.0069)	-0.0200*** (0.0070)	-0.0206*** (0.0070)	-0.0197*** (0.0071)	-0.0172** (0.0083)
CAR0to9	-0.0261*** (0.0094)	-0.0258*** (0.0097)	-0.0263*** (0.0095)	-0.0223** (0.0100)	-0.0276** (0.0123)	-0.0201 (0.0122)	-0.0199 (0.0123)	-0.0177 (0.0120)	-0.0122 (0.0123)	-0.0234 (0.0143)	-0.0237*** (0.0075)	-0.0235*** (0.0077)	-0.0249*** (0.0076)	-0.0232*** (0.0079)	-0.0193** (0.0096)
CAR0to10	-0.0248** (0.0097)	-0.0239** (0.0100)	-0.0257** (0.0099)	-0.0216** (0.0104)	-0.0252* (0.0128)	-0.0202 (0.0125)	-0.0191 (0.0128)	-0.0192 (0.0125)	-0.0136 (0.0127)	-0.0216 (0.0152)	-0.0201** (0.0079)	-0.0195** (0.0080)	-0.0222*** (0.0080)	-0.0202** (0.0084)	-0.0089 (0.0100)
Size and Leverage	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Headquarter FE		X					X					X			
Year FE			X					X					X		
Event FE				X					X					X	
Company FE					X					X					X
Observations	3877	3877	3875	3870	3799	2702	2702	2700	2697	2613	4107	4107	4105	4100	4020
Clusters	142	142	140	135	140	112	112	110	107	111	153	153	151	147	152

Notes: Robust standard errors are clustered on the event-level. Standard errors in parentheses: *p<0.1, ** p<0.5, *** p<0.01.

Table D.5: The effect of assassination events on stock price - Admin1 control sample.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
CAR0to0	-0.0070 (0.0046)	-0.0067 (0.0047)	-0.0043 (0.0039)	-0.0057 (0.0043)	-0.0116 (0.0076)	-0.0073 (0.0058)	-0.0078 (0.0067)	-0.0046 (0.0053)	-0.0037 (0.0055)	-0.0155 (0.0103)
CAR0to1	-0.0115 (0.0080)	-0.0120 (0.0086)	-0.0082 (0.0075)	-0.0103 (0.0085)	-0.0295* (0.0172)	-0.0071 (0.0101)	-0.0092 (0.0117)	-0.0029 (0.0103)	-0.0031 (0.0116)	-0.0275 (0.0214)
CAR0to2	-0.0132 (0.0098)	-0.0134 (0.0107)	-0.0090 (0.0096)	-0.0099 (0.0106)	-0.0192 (0.0179)	-0.0054 (0.0118)	-0.0078 (0.0138)	0.0006 (0.0127)	-0.0002 (0.0132)	-0.0103 (0.0226)
CAR0to3	-0.0105 (0.0108)	-0.0108 (0.0117)	-0.0061 (0.0100)	-0.0085 (0.0115)	-0.0202 (0.0185)	-0.0011 (0.0125)	-0.0016 (0.0144)	0.0059 (0.0127)	0.0055 (0.0143)	-0.0113 (0.0220)
CAR0to4	-0.0170 (0.0108)	-0.0160 (0.0118)	-0.0126 (0.0103)	-0.0148 (0.0116)	-0.0269 (0.0189)	-0.0096 (0.0124)	-0.0086 (0.0143)	-0.0021 (0.0128)	-0.0008 (0.0144)	-0.0198 (0.0222)
CAR0to5	-0.0147 (0.0111)	-0.0126 (0.0122)	-0.0117 (0.0114)	-0.0120 (0.0124)	-0.0279 (0.0216)	-0.0064 (0.0120)	-0.0044 (0.0138)	-0.0001 (0.0128)	0.0065 (0.0140)	-0.0241 (0.0251)
CAR0to6	-0.0163 (0.0117)	-0.0151 (0.0128)	-0.0130 (0.0121)	-0.0117 (0.0126)	-0.0180 (0.0221)	-0.0092 (0.0132)	-0.0080 (0.0148)	-0.0037 (0.0138)	0.0037 (0.0146)	-0.0115 (0.0246)
CAR0to7	-0.0193* (0.0112)	-0.0162 (0.0127)	-0.0164 (0.0119)	-0.0127 (0.0126)	-0.0233 (0.0233)	-0.0105 (0.0131)	-0.0059 (0.0147)	-0.0044 (0.0137)	0.0056 (0.0139)	-0.0170 (0.0253)
CAR0to8	-0.0214* (0.0125)	-0.0187 (0.0140)	-0.0177 (0.0131)	-0.0128 (0.0136)	-0.0302 (0.0254)	-0.0140 (0.0138)	-0.0107 (0.0157)	-0.0078 (0.0144)	0.0044 (0.0145)	-0.0218 (0.0270)
CAR0to9	-0.0267** (0.0131)	-0.0207 (0.0148)	-0.0235* (0.0136)	-0.0129 (0.0134)	-0.0339 (0.0273)	-0.0206 (0.0154)	-0.0131 (0.0171)	-0.0156 (0.0161)	0.0010 (0.0152)	-0.0263 (0.0290)
CAR0to10	-0.0281** (0.0136)	-0.0215 (0.0152)	-0.0250* (0.0143)	-0.0139 (0.0139)	-0.0308 (0.0275)	-0.0195 (0.0159)	-0.0103 (0.0174)	-0.0145 (0.0164)	0.0075 (0.0152)	-0.0227 (0.0301)
Size and Leverage	X	X	X	X	X	X	X	X	X	X
Profitability						X	X	X	X	X
Cubic Terms						X	X	X	X	X
Headquarter FE		X					X			
Year FE			X							
Event FE				X					X	
Company FE					X					X
Observations	676	675	676	673	605	658	657	658	653	586
Clusters	92	92	92	89	89	92	92	92	87	88

Notes: Robust standard errors are clustered on the event-level. Standard errors in parentheses: *p<0.1, ** p<0.5, *** p<0.01.

Table D.6: The effect of assassination events on stock prices without company ties.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
CAR0to0	-0.0050 (0.0061)	-0.0049 (0.0064)	-0.0028 (0.0046)	-0.0058 (0.0048)	-0.0034 (0.0072)	-0.0041 (0.0060)	-0.0050 (0.0063)	-0.0023 (0.0045)	-0.0059 (0.0045)	-0.0029 (0.0074)
CAR0to1	-0.0019 (0.0040)	-0.0025 (0.0042)	-0.0022 (0.0048)	-0.0035 (0.0075)	-0.0047 (0.0050)	-0.0018 (0.0040)	-0.0022 (0.0042)	-0.0023 (0.0045)	-0.0046 (0.0071)	-0.0052 (0.0051)
CAR0to2	0.0002 (0.0052)	-0.0004 (0.0057)	-0.0005 (0.0062)	-0.0012 (0.0087)	-0.0070 (0.0080)	-0.0009 (0.0051)	-0.0026 (0.0058)	-0.0012 (0.0056)	-0.0032 (0.0079)	-0.0086 (0.0084)
CAR0to3	-0.0014 (0.0074)	-0.0026 (0.0078)	-0.0009 (0.0072)	-0.0016 (0.0094)	-0.0102 (0.0102)	-0.0074 (0.0072)	-0.0109 (0.0077)	-0.0061 (0.0066)	-0.0077 (0.0086)	-0.0108 (0.0107)
CAR0to4	-0.0008 (0.0067)	-0.0016 (0.0067)	-0.0029 (0.0075)	-0.0006 (0.0101)	-0.0136 (0.0096)	-0.0066 (0.0067)	-0.0097 (0.0069)	-0.0082 (0.0069)	-0.0055 (0.0095)	-0.0135 (0.0099)
CAR0to5	0.0007 (0.0078)	-0.0020 (0.0081)	-0.0031 (0.0085)	-0.0048 (0.0117)	-0.0132 (0.0106)	-0.0041 (0.0082)	-0.0094 (0.0086)	-0.0080 (0.0082)	-0.0102 (0.0110)	-0.0129 (0.0110)
CAR0to6	-0.0013 (0.0094)	-0.0052 (0.0095)	-0.0072 (0.0102)	-0.0148 (0.0134)	-0.0088 (0.0119)	-0.0069 (0.0094)	-0.0133 (0.0095)	-0.0127 (0.0100)	-0.0223* (0.0128)	-0.0093 (0.0124)
CAR0to7	0.0031 (0.0119)	0.0001 (0.0118)	-0.0009 (0.0111)	-0.0096 (0.0124)	-0.0007 (0.0156)	-0.0020 (0.0123)	-0.0068 (0.0121)	-0.0065 (0.0114)	-0.0169 (0.0118)	-0.0013 (0.0164)
CAR0to8	-0.0019 (0.0120)	-0.0039 (0.0117)	-0.0064 (0.0111)	-0.0195 (0.0132)	-0.0066 (0.0166)	-0.0058 (0.0120)	-0.0100 (0.0116)	-0.0111 (0.0111)	-0.0268** (0.0122)	-0.0066 (0.0171)
CAR0to9	-0.0003 (0.0137)	-0.0043 (0.0136)	-0.0047 (0.0126)	-0.0151 (0.0144)	-0.0100 (0.0172)	-0.0058 (0.0138)	-0.0121 (0.0134)	-0.0106 (0.0122)	-0.0233* (0.0130)	-0.0101 (0.0177)
CAR0to10	-0.0047 (0.0174)	-0.0059 (0.0176)	-0.0058 (0.0151)	-0.0134 (0.0166)	-0.0164 (0.0193)	-0.0093 (0.0171)	-0.0129 (0.0170)	-0.0110 (0.0146)	-0.0197 (0.0153)	-0.0168 (0.0198)
Size and Leverage	X	X	X	X	X	X	X	X	X	X
Profitability						X	X	X	X	X
Cubic Terms						X	X	X	X	X
Headquarter FE		X					X			
Year FE			X							
Event FE				X					X	
Company FE					X					X
Observations	1472	1470	1472	1471	1403	1434	1432	1434	1433	1368
Clusters	62	62	62	61	62	62	62	62	61	62

Notes: Robust standard errors are clustered on the event-level. Standard errors in parentheses: *p<0.1, ** p<0.5, *** p<0.01.

Table D.7: The Effect of Assassination Events on Institutional Investor Holdings - Baseline Event Study Companies Only

Dep. Variable:	Institutional Investor Type							
	IO	Banks	Insurance Comp.	Investment Comp.	Investment Advisors	Pesion Funds & Endowment	Hedge Funds	Top 5
Assassination	0.0001 (0.0050)	0.0000 (0.0001)	-0.0001 (0.0001)	-0.0011 (0.0018)	0.0054 (0.0045)	-0.0005 (0.0006)	-0.0036*** (0.0011)	0.0061 (0.0043)
Size and Leverage	X	X	X	X	X	X	X	X
Company FE	X	X	X	X	X	X	X	X
Company-specific quadratic time trend	X	X	X	X	X	X	X	X
R-squared	0.717	0.516	0.591	0.601	0.660	0.615	0.410	0.523
Observations	3042	3042	3042	3042	3042	3042	3042	3042
Mean	0.2575	0.0003	0.0006	0.0573	0.1658	0.0137	0.0197	0.1098

Notes: Robust standard errors clustered at the company-level in parentheses: *p<0.1, ** p<0.05, *** p<0.01.

Table D.8: The Relationship between Assassinations and the Change in Tax Revenue Shares

	Dependent Variable: Δ Tax Share			
	(1)	(2)	(3)	(4)
Assassination	0.012 (0.023)	0.011 (0.023)	0.012 (0.024)	0.013 (0.025)
Country FE		X	X	
Year FE			X	
Country $\times$ Year FE				X
R-squared	0.001	0.002	0.011	0.053
Observations	784	784	784	784

Notes: Robust standard errors in parentheses: *p<0.1, ** p<0.05, *** p<0.01.